Chapter 12
Destructive Mechanisms in Power Devices

This chapter will deal with some destructive mechanisms in power devices, and typical failure pictures for them will be shown. Failure analysis requires a lot of experience, especially regarding the conditions in the power circuit at failure, which must be carefully considered. Although some of the failure pictures appear to be similar, it is difficult to draw conclusions only from pictures. However in practice the engineer often has the problem to find the reason for failures, and the following sections might be helpful.

12.1 Thermal Breakdown – Failures by Excess Temperature

In Chap. 2 the intrinsic carrier density n_i was explained, it strongly depends on the temperature as described in Eq. (2.6). In silicon, n_i amounts approx. to 10^{10} cm^{-3} at room temperature, and it is negligible compared to the background doping. However, n_i increases rapidly with increasing temperature. Therefore, at very high temperatures, the thermal generation becomes the dominant mechanism for the creation of carriers.

With the introduction of an intrinsic temperature T_{int} similar to [Gha77], one can estimate, when the rise of some critical mechanisms in a device with increasing temperature can be expected. T_{int} is that temperature, at which the density of carriers n_i generated by thermal generation is equal to the background doping N_D. It is drawn in Fig. 12.1 as a function of N_D. Below T_{int} the carrier density is only weakly dependent on temperature. Above T_{int} the carrier density increases exponentially with temperature according to Eq. (2.6). From Fig. 12.1 we can see that for a high-voltage device, which requires an N_D in the range of 10^{13} cm^{-3}, T_{int} will be reached at a much lower temperature compared to a device with a lower voltage rating, at which, e.g. a range of 10^{14} cm^{-3} is used for N_D.

However, this point of view is much too simplified. The intrinsic carrier density is defined for thermal equilibrium, while a device in application is usually never operating in thermal equilibrium. Therefore, one must also consider in which operation mode and by which effect an increased or locally increased temperature is reached.

For low-voltage MOSFETs operated in a short time interval in avalanche breakdown, it was found that destruction occurs if the temperature increases just below

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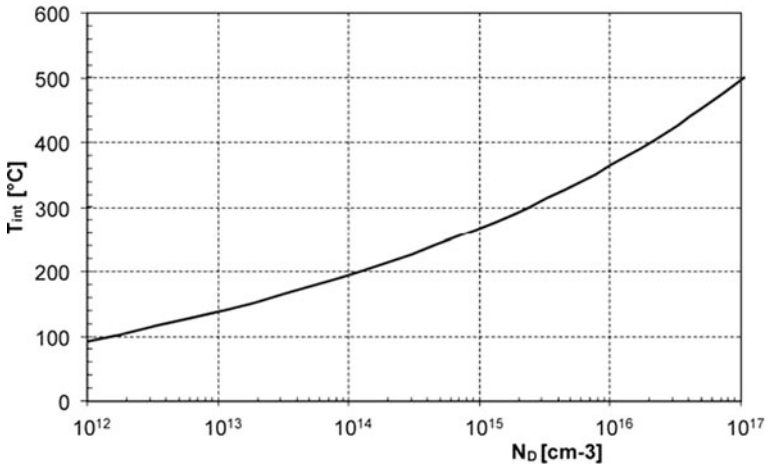


Fig. 12.1 Intrinsic temperature in silicon as a function of background doping

the temperature T_{int}, which was in the range 320°C for the used 60 V MOSFET [Ron97]. This agrees with Fig. 12.1, if the doping N_{D} of 4×10^{15} cm^{-3} is assumed which is reasonable for this voltage range.

If a bipolar device is in the forward conduction mode, e.g., in a surge current event, it is flooded with free carriers in a density in the range above 10^{16}, even slightly above 10^{17} cm^{-3} can occur. When the thermal generation amounts to a carrier density in this range, it becomes the dominating mechanism. Therefore, in such short time events, very high temperatures may occur without a failure in the device. In the surge current mode, a T_{int} up to the range of 500°C is to be expected.

In the blocking mode, a space charge region builds up and carriers are removed from the depleted zone. Their density is given by the leakage current, which is low for most modern power devices (except for gold-diffused devices). Thermal stability is now determined by the height of the leakage current.

Therefore the associated mechanism which leads to a high temperature must be taken into account. For the investigation of stability, one has to consider the temperature dependency of the electrical mechanism which leads to high losses and therewith high temperature.

If the heating is caused by a high leakage current, the leakage current will further increase in regions of high temperature. These regions will get hotter, which again leads to more increased leakage current. Such behavior shall be termed here as a *positive feedback*. If the high power loss density cannot be extracted by the cooling of the device, the device will be destroyed inevitably.

If the high losses are generated by a high voltage above the breakdown voltage V_{BD}, which drives the device in the avalanche breakdown, losses are created and the temperature increases. However, with increasing temperature, V_{BD} increases. The region, where avalanche breakdown occurs, will move to the regions of the device where the temperature is lower. Even if such electrical mechanisms lead to local

filaments, the increase of temperature releases the local stress, leading to a *negative feedback effect*.

However, if the temperature reaches T_{int}, the thermal generation will become the dominating effect, and then the temperature increase acts as a positive feedback. Inhomogenities in current density, even if they are small, will be amplified rapidly. If the temperature reaches T_{int}, one has to expect current tubes or filaments, considering a device with an area above some square millimeter.

If P_{gen} is the generated power density and P_{out} is the power density that can be maximally be drawn out via the package and heat sink, one can express a condition for thermal runaway [Lin08]:

$$\frac{\partial P_{gen}}{\partial T} > \frac{\partial P_{out}}{\partial T} \tag{12.1}$$

If this condition is fulfilled for a stationary operation point, a fast exponential temperature increase will occur. Equation (12.1) is a general form of an equation that was used for thermal stability of bipolar transistors [How74]:

$$S = R_{th} \cdot V_C \frac{\partial I_C}{\partial T} \tag{12.1a}$$

If the stability factor S increases > 1, any temperature disturbance will grow and thermal runaway will occur.

Finally, the destruction of the device is always due to high temperature. Failed devices show local regions of molten semiconductor material. If the temperature increase occurs locally, in a very small point-like area of the device, one can find even cracks in the crystal lattice. However, it must be distinguished by which effect the temperature increase was generated. Some of these effects will be described below.

As a simple example, Fig. 12.2 shows an IGBT device, which has failed due to very high power losses. The IGBT was only stressed with forward current and the failure was caused by a too low gate voltage V_G. One can recognize a comparatively large molten area (>1 mm^2) at the emitter side. It is located typically close to the center of the device and close to the bond wires.

If overtemperature occurs in an application in which the device is switched between the blocking mode and the conducting mode at a high frequency, the destruction picture may be different. With increased temperature, the blocking capability is lost first. In almost all devices with planar junction termination the breakdown will occur at the edge. Therefore the point of destruction will be at the edge of the device or at least a small part of the edge should be included.

12.2 Surge Current

In the application of a diode or a thyristor in a rectifier, momentary high over-current pulses can occur. Therefore the possible surge current is determined and given in the datasheets for rectifier diodes, fast diodes, and thyristors. During the qualification

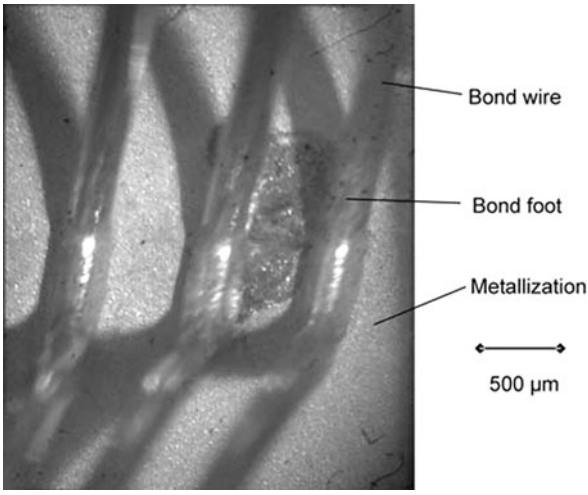


Fig. 12.2 IGBT-Chip destroyed by excess temperature

of a diode, a single sinus half-wave of the grid current is imposed in the forward direction of the diode. Figure 12.3 shows the surge current measurement of a fast 1200 V diode with an area of 7×7 mm^2, the waveforms of current, voltage, and additionally the power $p = v \cdot i$ are pictured as a function of time.

Due to the junction voltage of the measured diode and the junction voltage of a thyristor in the measurement equipment, the current pulse duration is not 10 ms as it should be for a grid frequency of 50 Hz, but it is 7.5 ms. In Fig. 12.4 the *I–V*

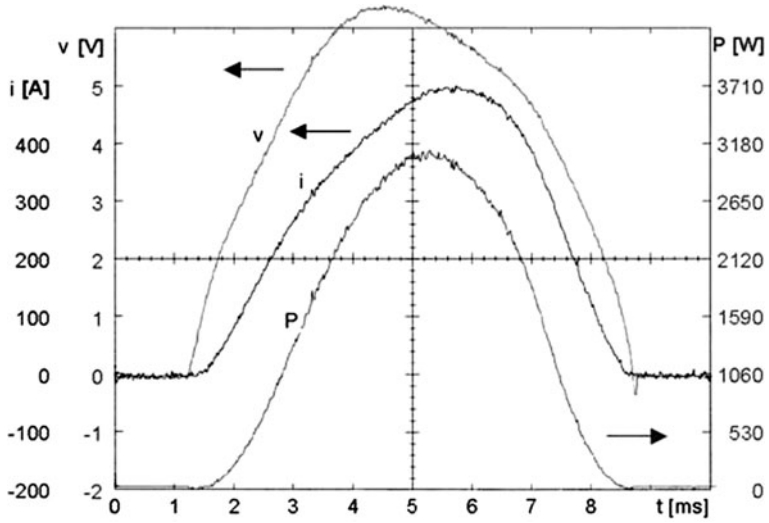


Fig. 12.3 Surge current load of a fast diode. Voltage, current, and power depending on time

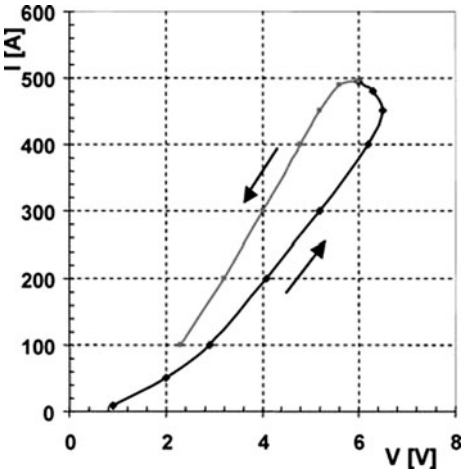


Fig. 12.4 *I–V* characteristic of the surge current load in Fig. 12.3

characteristic from the measurement in Fig. 12.3 is shown. Since a high temperature is reached in the device, the characteristic splits into an ascending and descending branch. In the descending branch the voltage drop is significantly lower.

For the dissipated power in Fig. 12.3 the temperature in the semiconductor is estimated in Fig. 12.5 with a thermal simulation using the simulator SIMPLORER. The diode is packaged as shown in Fig. 11.13, the thickness of the Al_2O_3 ceramic is 0.63 mm. The heat flux is fed in the volume of the low-doped n^--layer of the device in the form of a sinus-square function within a time of 7.5 ms and amplitude of 3060 W, according to the measurement in Fig. 12.3. The temperature dependency

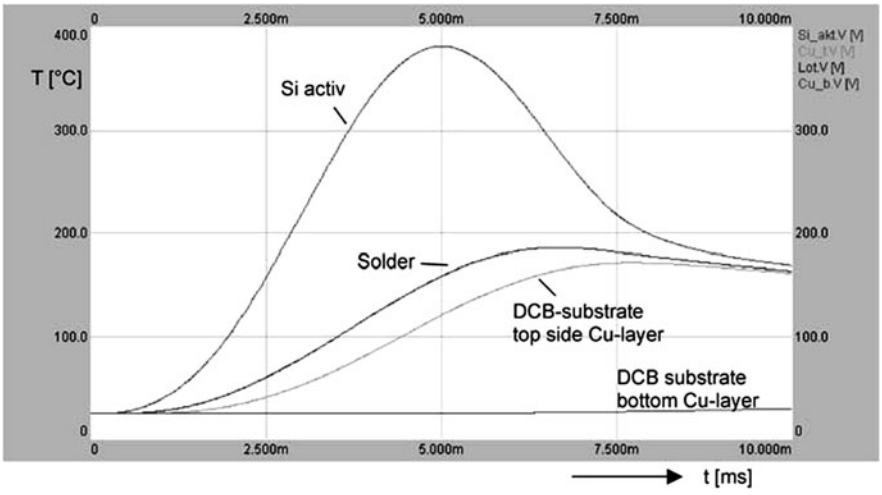


Fig. 12.5 Simulation of temperatures for the surge current event in Fig. 12.3

of thermal conductivity in silicon is considered according to Eq. (11.5), since this will be significant for the result at the expected high temperature increase. In the estimation in Fig. 12.5, the temperature in the n^--layer (Si-active) of the device increases up to 382°C.

This high temperature may explain the strong difference of the forward voltage in Fig. 12.4 in the descending branch compared to the ascending branch. The voltage drop V_F consists of $V_F = V_j + V_{drift} \cdot I_F$ decreases at high temperature because n_i is strongly increasing with temperature. For V_{drift}, the temperature dependency can be discussed using Eq. (5.47)

$$V_{drift} = \frac{w_B^2}{(\mu_n + \mu_p) \cdot \tau_{eff}} \tag{12.2}$$

where τ_{eff} contains the carrier lifetime τ_p as well as the emitter influence, see Eq. (5.52).

To this temperature dependency contribute the following effects:

- The carrier lifetime. It increases with temperature. This leads to a decrease of the forward voltage drop with temperature.
- The emitter recombination: For this, Auger recombination is important and at a high current density τ_{eff} will become very small. Additionally, in many modern devices the emitter depth is smaller than the diffusion length and then the emitter depth must be used in the emitter parameter. However, no strong temperature dependency of emitter recombination is to be expected.
- The mobilities. They decrease strongly with temperature. This effect leads to an increase of the forward voltage.
- The temperature dependency of the resistance of metallization and bond wires. This resistance increases with temperature.
- Finally the density of thermal-generated carriers n_i depends strongly on temperature. If n_i significantly contributes to the density of free carriers – this is in the range of 10^{17} cm^{-3} for surge current events – then a significant decrease of the forward voltage is to be expected. Now Eq. (12.2) is no longer valid. However, Eq. (5.34) which was used to derive the basics of forward conduction can be written as

$$V_{drift} = \frac{j}{q} \int_0^{w_B} \frac{1}{\left(\mu_n(x) + \mu_p(x)\right) p(x)} \mathrm{d}x \tag{12.3}$$

The density of free carriers $p(x) \approx n(x)$ is strongly increasing and V_{drift} is decreasing.

The form of the *I*–*V* characteristic will therefore be very different for different diode fabrication technologies. The behavior like Fig. 12.4 is normally observed in some special fast diodes. Finally, the failure can be caused by the following mechanisms:

(a) Melting of the top side metallization. This occurs especially in bonded diodes in power modules.
(b) Mechanical destruction, cracks in the device, caused by very high temperature and mechanical stress resulting from thermal expansion.
(c) If finally n_i is dominating, the characteristic behavior of a resistor with a negative temperature coefficient occurs [Sil73]. The forward voltage decreases strongly. A positive feedback occurs. A filamentation of the current into tubes with very high current density is to be expected.

In [Sil73] it is estimated that the negative temperature coefficient of the resistance according to mechanism (c) occurs after the temperature has increased to an amount, at which n_i approximates to 0.3 $\overline{n}$. There are some hints that there is a dependency on the device area. Small diodes can bear a higher current density since there are fewer possibilities for the formation of filaments. If the failure occurs according to (c), then it will be typically close to the edge of the active area, since these are the locations with the highest current densities.

For wire-bonded diodes in IGBT modules, as shown in Fig. 11.13, the anode layer with a typical low junction depth is on the top. The heat dissipating volume is close to the metallization layer and bond wires. For such diodes, failure according to mechanism (a) is expected. In a more detailed simulation with the Sentaurus$_{TCAD}$ device simulator [Syn07], it was found that the metallization layer and the bond foot arrangement are of high influence to the occurring temperatures and the surge current capability [Hei08b]. The thicker the metallization the better is the capability to absorb the heat especially in short times, since it acts as additional thermal capacitance. Another important factor on the surge current capability of diodes is the location and size of the contact area of the bond wires. A high ratio of bond foot area to the diodes anode will increase the surge current capability.

A surge current stress below the destruction limit will not lead to irreversible modifications in the semiconductor itself. If the surge current is increased above the value in Fig. 12.4, the split of the characteristics will increase and higher temperature will occur in the device. Finally destruction of the device occurs.

However, already in the simplified temperature estimation in Fig. 12.5 the temperature in the chip solder layer grows up to 186°C. Such a temperature is already close to thermal softening of solder layers. Therefore irreversible modifications in solder layers, and also in metallization and bond wires, may occur. The surge current capability therefore is intended for singular overload events and it is not intended for regular operation of a power semiconductor.

The temperature in the bottom Cu-layer of the DCB substrate has only grown up to 27°C, this is negligible. Therefore the influence of further components of the package can be neglected at surge current conditions; hence all effects happen in the semiconductor and in the immediately adjacent layers.

The surge current capability of a fast diode is typically at 10 – 12 times of the rated current. The surge current capability of a diode for grid frequency operation or of a thyristor is typically in the range of 20 times of the rated current, since these

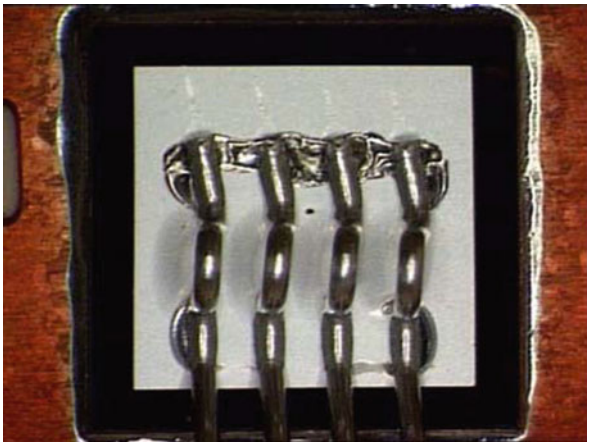


Fig. 12.6 Wire-bonded diode, destroyed by surge current

devices are manufactured with high carrier lifetime, and the forward voltage drop is lower.

A diode which failed due to surge current is shown in Fig. 12.6. The molten regions close to the bond feet are typical. The hottest surface spot at surge current is beside the bond feet [Hei08b]. The failure mechanism here is according to (a). The molten area for this case is always in the active area of the device. Such pictures allow a clear identification as a surge current fault during analysis in the quality department of the device manufacturer.

The failure in an application must not be during a sinus-shaped pulse. The current pulse might be of a different shape. A general description is done by the i^2t value in data sheets, which holds for arbitrary current pulses. The failures are due to exorbitant heating of the device by too high currents and the occurrence of one of the described mechanisms.

12.3 Overvoltage – Voltage Above Blocking Capability

The blocking capability of power devices is limited by avalanche breakdown. Avalanche breakdown occurs above the rated voltage of the device. Most power devices can sustain some current in the avalanche breakdown mode. However, the data sheet of the manufacturer excludes operation of the device in the avalanche mode, if the device is not avalanche rated.

Several MOSFETs and diodes in the range up to 1000 V are avalanche rated; this allows short-time operation in the avalanche mode. The maximum dissipated energy in the avalanche E_{av} is specified in the data sheets of the manufacturers, in general form it is given by :

$$E_{\mathrm{av}} = \int_{t_{\mathrm{av}}} V_{\mathrm{BD}} \cdot i(t)\mathrm{d}t \tag{12.4}$$

where V_{BD} is the device breakdown voltage, $i(t)$ the current pulse in avalanche, and t_{av} is the time of the current pulse. Usually avalanche occurs at "unclamped inductive switching," where the MOSFET is turned off in a circuit with an inductor L in series. The voltage now rises up to the breakdown voltage and the current decreases with

$$\frac{di}{dt} = \frac{V_{BD} - V_{bat}}{L} \tag{12.5}$$

If $i(t)$ decays linearly from $I_{av(peak)}$ to zero during t_{av}, as it is the case when the energy of an inductor is dissipated, Eq. (12.4) can be expressed as

$$E_{av} = \frac{1}{2} \cdot V_{BD} \cdot I_{av(peak)} \cdot t_{av} \tag{12.6}$$

and the dissipated energy in this case is the stored energy in the inductor $0.5 \cdot L \cdot I^2_{av(peak)}$ and additionally the energy $0.5 \cdot V_{bat} \cdot I_{av(peak)} \cdot t_{av}$ which is delivered by the voltage source V_{bat} during t_{av}. Avalanche rating of low-voltage MOSFETs can be up to $E_{av} = 1$ J, such energy can only be dissipated in single events and never in continuous mode at a high operation frequency.

The design of the MOSFETs for avalanche capability is in a way, that the breakdown occurs in the volume of the device, e.g., in a planar MOSFET at the p^+-layer in the center of the cell (Fig. 9.4) where the n^--base is narrowest. Avalanche capability is also possible for trench MOSFETs (Fig. 9.6), for these structures an additional design effort is necessary to avoid the location of avalanche at the trench corners [Kin05].

The occurrence of breakdown in the volume of the device is also possible for diodes with a beveled junction termination of positive angle (Fig. 4.17). The breakdown will occur at the edge for diodes with planar junction terminations with floating potential rings (Fig. 4.19). Well-designed potential rings can also bear current in the avalanche mode, even if it flows mainly at the edge. However, above 1200 V, one can only rarely find avalanche rating and if it is with more restrictive conditions.

For avalanche capability, as second condition is that branches with negative differential resistance (NDR) must be avoided. This will be explained more in detail in the following.

For a higher rated blocking voltage the device must have a lower doping, see Fig. 3.17. If avalanche occurs, the shape of the electric field is triangular or trapezoidal and electron–hole pairs are generated in the region with high electric field. The generation depends exponentially on the electric field and the main part is generated close to the pn-junction.

Holes are flowing to the anode and electrons to the cathode. Within the space charge, we can include the generated carriers p_{av} and n_{av} in the basic equation (2.90) according to their polarity, in one-dimensional expression as

$$\frac{dE}{dx} = \frac{q}{\varepsilon_0 \cdot \varepsilon_r}(N_{D^+} + p_{av} - n_{av}) \tag{12.7}$$

At the anode side border of the space charge, no electrons arrive and the reverse current is carried only by holes. We neglect the small contributions of diffusion current and recombination center-induced leakage current. Then at this position holds

$$p_{\mathrm{av}} = \frac{j_{\mathrm{R}}}{q \cdot v_{\mathrm{sat}}(p)} \tag{12.8}$$

where j_{R} denotes the reverse current and $v_{\mathrm{sat}}(p)$ denotes the saturation velocity of holes since at the given high electric field. At the cathode-side border the arriving reverse current is pure electron current. The density of generated electrons is

$$n_{\mathrm{av}} = \frac{j_{\mathrm{R}}}{q \cdot v_{\mathrm{sat}}(n)} \tag{12.9}$$

with $v_{\mathrm{sat}}(n)$ of 1.05×10^7 cm/s at $T = 300$ K.

The generated free carriers influence the effective doping N_{eff} and thereby the shape of the electric field. The feedback is shown with device simulation in Fig. 12.7. At the left-hand side, and at the pn-junction, it holds $N_{\mathrm{eff}} = N_{\mathrm{D}} + p_{\mathrm{av}}$. With increasing j_{R} the gradient of the electric field becomes more steep according to Eqs. (12.7) and (12.8) and at the pn-junction occurs an increased field peak.

At the nn$^+$-side, N_{eff} is lowered since $N_{\mathrm{eff}} = N_{\mathrm{D}} - n_{\mathrm{av}}$. If n_{av} is in the range of N_{D}, a part of the positively charged donors will be compensated. If we assume a high-voltage diode with N_{D} of 1.1×10^{13} cm^{-3} as in Fig. 12.7a, only a current density $j_{\mathrm{R}} = 19$ A/cm^2 is necessary to compensate the background doping for this design (a) completely. The gradient dE/dx becomes flat. In Fig. 12.7b the same conditions are taken, but the doping N_{D} is increased to 1.7×10^{13} cm^{-3} for this design (b).

The voltage corresponds to the area below $E(x)$. For design (a) the negative voltage area is always larger than the positive voltage area that occurs at the nn$^+$-junction. For design (b), up to a current density of 40 A/cm^2 there is a larger positive voltage area at the nn$^+$-junction, resulting in a positive branch of the post-avalanche characteristics.

In the simulated I–V characteristic, branches with positive and with negative differential resistance (NDR) occur at increased carrier density in avalanche, depending on the respective design [Hei05]. Figure 12.8 shows the post-avalanche behavior. Diode design (a) with very low background doping $N_{\mathrm{D}} = 1.1 \times 10^{13}$ cm^{-3} shows NDR already at 0.1 A/cm^2. Diode design (b) with increased doping shows as consequence a lower breakdown voltage, but then follows a branch with positive differential resistance.

For a further increased current density, a second voltage peak builds up at the nn$^+$-junction. Finally, at a very high avalanche current density, design (b) runs at the same line as (a), since at this condition the blocking capability is no longer determined by the doping, but only by the free carriers and the device thickness. Design (c) in Fig. 12.8 with thicker base has an extended branch of positive differential resistance.

The occurrence of branches with the NDR was first explained by Egawa [Ega66]. They are in conjunction with a hammock-like field shape, as to be seen in Fig. 12.7a

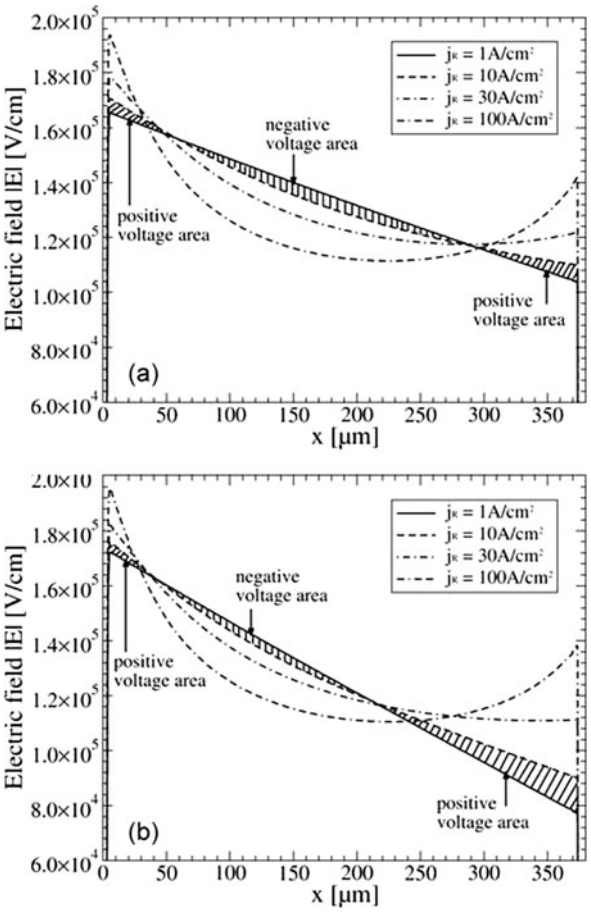


Fig. 12.7 Electric field $|E|$ at low and at increased avalanche current. (a) Design with $N_D = 1.1 \times 10^{13}$ cm^{-3}, negative differential resistance, (b) design with $N_D = 1.7 \times 10^{13}$ cm^{-3}, positive differential resistance up to some 10 A/cm^2. Fig from [Lut09] © 2009 IEEE

and b for high current density. This shall be called Egawa-type field. Such fields and branches with NDR have the precondition that the electron current n_{av}, calculated with Eq. (12.9), is higher than N_D. This condition is fulfilled for low-voltage devices only at very high current densities, but it may be reached at a moderate current density in high-voltage devices with low background dopings. This is one of the reasons why high-voltage devices are usually not avalanche rated.

At the nn$^+$-junction, avalanche is triggered by electrons and we have to use the multiplication factors for electrons, which are much higher than that for holes, see Fig. 3.15. Therefore impact ionization will occur at the nn$^+$-junction already at a lower electric field. Impact ionization at the nn$^+$-junction will create electron–hole pairs, the holes are flowing to the pn-junction and will increase the avalanche at the pn-junction. A positive feedback between impact ionization at the left and the right side will occur. Avalanche at an nn$^+$-junction has been described as a failure mechanism of devices in [Ega66, How70].

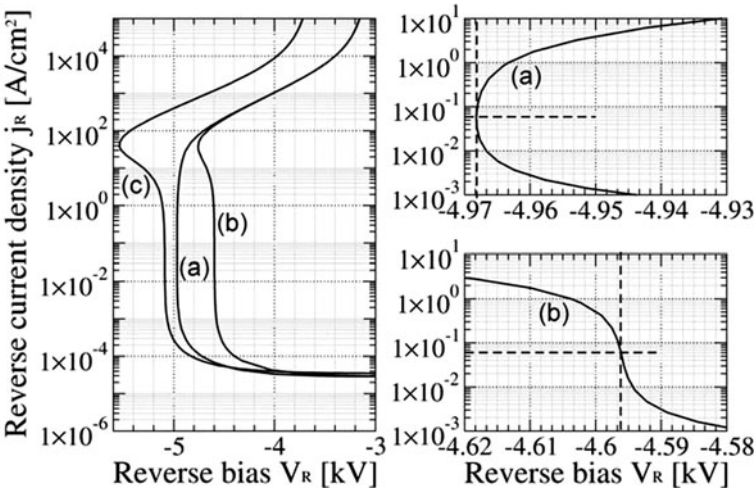


Fig. 12.8 Influence of base doping and base width to the static avalanche characteristics. (**a**) and (**b**) with $w_B = 375\,\mu m$ and different doping of 1.1×10^{13} cm^{-3} (**a**) and 1.7×10^{13} cm^{-3} (**b**), (**c**) with 1.7×10^{13} cm^{-3} and wider w_B of 450 µm. Fig from [Lut09] © 2009 IEEE

In Fig. 12.7, the difference between (a) and (b) is only the increased doping in (b). The range of positive differential resistance can be extended by a thicker base w_B . Most effective are buffer layers, in which a region of increased doping is arranged in front of the nn$^+$-layer. With special buffers, fields at the nn$^+$-junction can be limited and branches of negative differential resistance can be widely avoided. Details are given in [Fel06].

In devices with planar junction termination, the edge usually limits the maximal possible blocking capability[1]. Avalanche breakdown first occurs at the edge. Therefore current densities high enough to lead to Egawa-type fields can occur locally at the edge. For a failure by overvoltage it is typical that the edge of the device is included in the destroyed area. Figure 12.9 shows the failure picture of a 1700 V diode. The device has a planar junction termination with potential rings similar to Fig. 4.19. Three potential rings can be seen in the upper part of the figure. The point of destruction is located between the p-anode layer and the first potential ring. This is one of the positions of the highest electric field, as it is marked in Fig. 4.19.

The occurrence of such a failure position indicates that the failure was caused due to voltage. However, Fig. 12.9 does not allow a clear decision, whether an overvoltage above the rated voltage of the device was applied or the particular device had a weak point induced in the manufacturing process. A failure picture as in Fig. 12.9 is only then occurring, if no high current was flowing across the point of destruction.

[1]Exception MOSFETS. In MOSFETs the cell geometry is adjusted that the avalanche breakdown occurs first below the cells and not at the edge, see above.

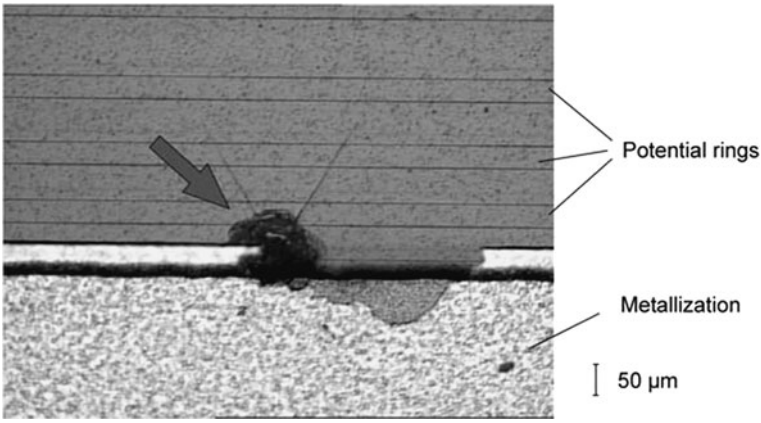


Fig. 12.9 1700-V diode destroyed by voltage

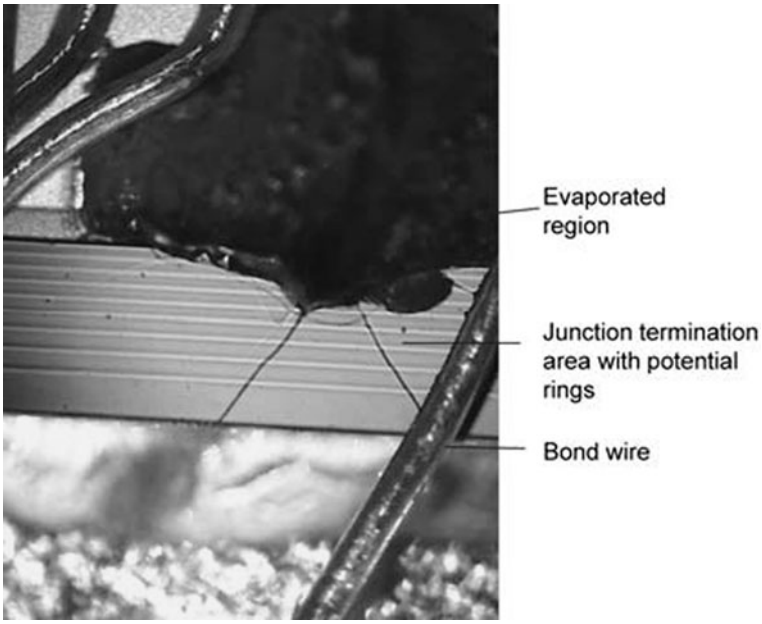


Fig. 12.10 3.3-kV diode possibly destroyed by overvoltage

A picture of a destroyed semiconductor, in which after failure a high current was flowing, is shown in Fig. 12.10. Part of the edge and a big part of the active area are evaporated. If such a picture occurs, one can presume that the destruction occurred at first at the edge and then it propagated toward the bond wires.

However, the cracks in the crystal lattice are not typical. These cracks indicate a local hot spot in a small, point-like position. Such failure pictures can also be

found at strong dynamic avalanche (dynamic avalanche of the third degree, see Sect. 12.4.2). Therefore this failure picture is not unequivocal.

12.4 Dynamic Avalanche

12.4.1 Dynamic Avalanche in Bipolar Devices

During switching of all bipolar devices the increase of the voltage occurs at an instant, at which a large part of the stored carriers, which have conducted the forward current before, is still present in the device. This stored charge is partially removed during the voltage increase, and it flows as hole current through the space charge region.

Figure 12.11 shows the process in a simplified way. The pn-junction at the position $x = 0$ represents the blocking pn-junction of a bipolar device. Between the junction and w_{SC} the space charge region has extended, for supporting the applied voltage. Between w_{SC} and the end of the lowly doped layer exists a plasma zone, in which $n \approx p$ holds. The effects at the right side shall be neglected in this first approximation. At this position there will be either an nn^+-junction of a diode, or the collector layer of an IGBT, or the anode layer of a GTO thyristor, etc. As long as very hard switching conditions are not applied, no space charge is build up at this position.

Through the space charge the current flows as hole current, $j = j_p$. The density of holes p can be calculated from the current density at this instant:

$$p = \frac{j}{q \cdot v_{\mathrm{sat(p)}}} \tag{12.10}$$

In this equation $v_{\mathrm{sat(p)}}$ is the saturation drift velocity of holes under the condition of high fields, it amounts in silicon to approximately 1×10^7 cm/s and is close to the saturation drift velocity of electrons $v_{\mathrm{sat(n)}}$. A current density j of 100 A/cm^2 leads to $p = 8.2 \times 10^{13}\,\mathrm{cm}^{-3}$, which is already in the order of the background doping of a bipolar 1200 V device. The hole density p can no longer be neglected.

Holes have the same polarity as the positively charged ionized donors, hence their density now adds to the background doping to an effective doping N_{eff}:

$$N_{\mathrm{eff}} = N_{\mathrm{D}} + p \tag{12.11}$$

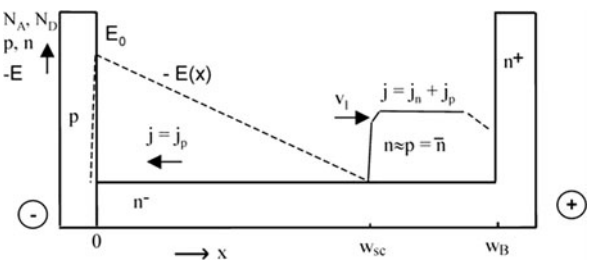


Fig. 12.11 Bipolar semiconductor device during the turn-off process

With the Poisson equation, N_{eff} determines the gradient of the electric field:

$$\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{q}{\varepsilon}\left(N_{\text{D}} + p\right) \tag{12.12}$$

so that $\mathrm{d}E/\mathrm{d}x$ is increased. With this, the field shape is steeper, E_0 is increased and the voltage, which drops across the space charge of width w_{SC}, is increased in the first instance. However, E_0 can rise only up to the avalanche field strength E_c. E_c will now be reached at an applied voltage far below the specified rated blocking voltage of the device and avalanche will set in. This process, which is now dominated by free carriers, is called dynamic avalanche.

This process occurs during turn-off of diodes, GTOs, and IGBTs, for further details, however, the specific peculiarities of the respective device and their physics must be considered.

12.4.2 Dynamic Avalanche in Fast Diodes

In the state which is drawn in Fig. 12.11, the shape of the electric field is approximately triangular, since no space charge can penetrate into the plasma layer. For a triangular field we can use the relation (3.84) between the avalanche breakdown voltage V_{BD} and doping:

$$V_{\text{BD}} = \frac{1}{2} \cdot \left(\frac{8}{B}\right)^{\frac{1}{4}} \cdot \left(\frac{q \cdot N_{\text{eff}}}{\varepsilon}\right)^{-\frac{3}{4}} \tag{12.13}$$

where for the ionization rates the proposal of Shields and Fulop is used with $B = 2.1 \times 10^{-35}\ \text{cm}^6/\text{V}^7$ and $n = 7$ at room temperature [Shi59, Ful67]. The derivation of this equation can be found in Chap. 3, Eqs. (3.75) to (3.84). This equation can be used since dynamic avalanche in this case is governed by holes.

If now Eqs. (12.10) and (12.11) are inserted in Eq. (12.13), we obtain a relation between the onset of dynamic avalanche and the current density in the space charge region, as shown in Fig. 12.12. For the background doping N_{D} the typical values for a 1700 V device and for a 3300 V device are used, $N_{\text{D}} = 4.3 \times 10^{13}$ and $1.7 \times 10^{13}\ \text{cm}^{-3}$, respectively.

From Fig. 12.12 one can recognize that for a device with high static blocking capability the avalanche onset voltage decreases strongly. In a 3.3 kV diode, dynamic avalanche sets in already at a reverse current density of 30 A/cm^2. For a reverse current density of 200 A/cm^2 the limit for the onset of dynamic avalanche has decreased down to 1050 V, and it is not much above the limit of a 1700 V device. There are only small differences between the two diodes voltage ratings, because the second term in Eq. (12.11) representing the free charge carriers is dominating.

Dynamic Avalanche of the First Degree

Close to the onset of dynamic avalanche, failures have been detected and assumed to be an unavoidable phenomenon of semiconductor physics [Por94]. Even a “silicon

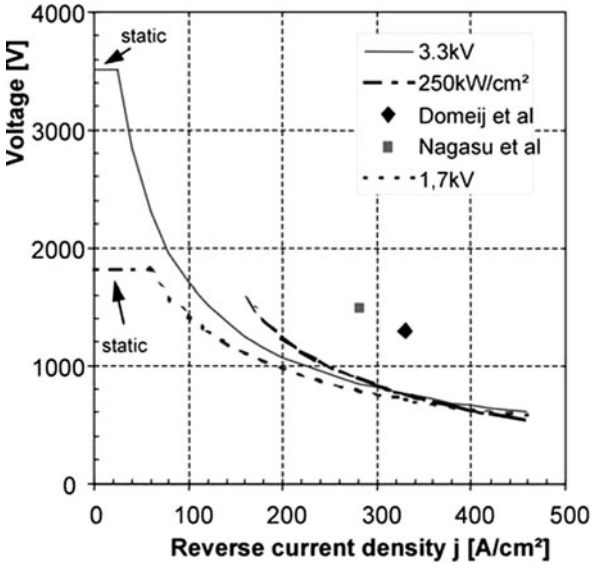


Fig. 12.12 Avalanche onset voltage depending on current density in reverse direction according to Eq. (12.13), former assumed limit of 250 kW/cm^2, reported operation points of 3.3-kV power diodes at max. power density of former work Domeij et al. [Dom99] and Nagasu et al. [Nag98] Reprinted from [Lut03] with permission from Elsevier

limit" at a power density $V{\cdot}I/A = 250\,\mathrm{kW/cm^2}$ was sometimes assumed [Sit02]. The line for 250 kW/cm^2 shown in Fig. 12.12 is close to the calculations with Eq. (12.13), the onset of dynamic avalanche at the pn-junction.

However, Schlangenotto [Sco89b] published another point of view. Moderate dynamic avalanche, dynamic avalanche of the first degree, should not be critical. Dynamic avalanche generates electrons in the space charge which counteracts the positive charge of holes and then holds

$$N_{\mathrm{eff}} = N_{\mathrm{D}} + p - n_{\mathrm{av}} \tag{12.14}$$

The increased hole density is partially compensated and this mechanism should be self-stabilizing. This was confirmed by diode designs bearing a significant amount of dynamic avalanche current [Lut97]. For 3.3 kV diodes Nagasu et al. [Nag98] and Domeij et al. [Dom99] reported measurements with 1.7 times higher voltages than the alleged limit. These results are also shown in Fig. 12.12.

The power loss density gives an estimation of the avalanche intensity, but it gives no physical explanation of a failure limit, since the high-power density occurs only during some 10 ns, while the dissipated energy is low and the temperature increase, if it is a one-time effect, is negligible.

However, for this to hold true it is a precondition that the device has no weak points in its design. In [Nag98] and [Tom96] it was shown that the edge of the active area is of great importance. In diodes with the common planar field limiting structures, the anode area is smaller than the cathode area. At the edge of the anode, additional current contributions from the n$^+$-region are occurring during forward conduction, as shown schematically in Fig. 12.13. The current density at the edge of

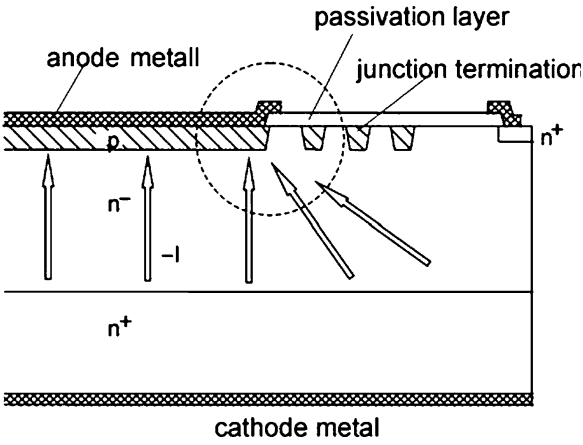


Fig. 12.13 Planar junction termination of a diode with increased current density at the edge of the active area. Reprinted from [Lut03] with permission from Elsevier

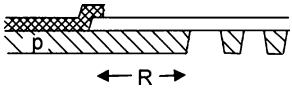


Fig. 12.14 Resistive layer for reduction of the current density at the edge of the anode. Reprinted from [Lut03] with permission from Elsevier

the anode is increasing and in device simulation one can observe a current filament already at the beginning of turn-off period. The implementation of a resistive zone at the edge, as shown in Fig. 12.14, is reducing this weak point. The p-layer is extended by a length R below the passivation layer. The anode side p-layers are only moderately doped in soft recovery diodes, therefore the region R acts as a pre-resistor for a current at the edge of the p-layer and the current density at the edge will be reduced. Such a structure at the edge was called "HiRC" structure (high reverse recovery capability) in [Mor00].

Dynamic Avalanche of the Second Degree

With an increasing current density, dynamic avalanche leads to filamentation of the current due to space charge effects as reported by Oetjen et al. [Oet00]. Figure 12.15 shows a static simulation of the electric field distribution and I–V characteristic of a p^+np^+ structure. This figure represents the process of Fig. 12.11; however, for an increased current density. The plasma layer of Fig. 12.11 is replaced by a p^+-layer which injects holes. Figure 12.15a displays the electric field at the current densities 500 and 1500 A/cm^2. The field has reached the avalanche field strength level at the pn-junction, in this figure at $x = 8\,\mu$m. The field shape for $j = 500$ A/cm^2 is approximately triangular. Avalanche generates electron–hole pairs in the region with high field strength. However, the generation is not local, since it happens within some length in the x-direction which is necessary for carrier acceleration. The holes are flowing to the left-hand side; the electrons are flowing to the right-hand side.

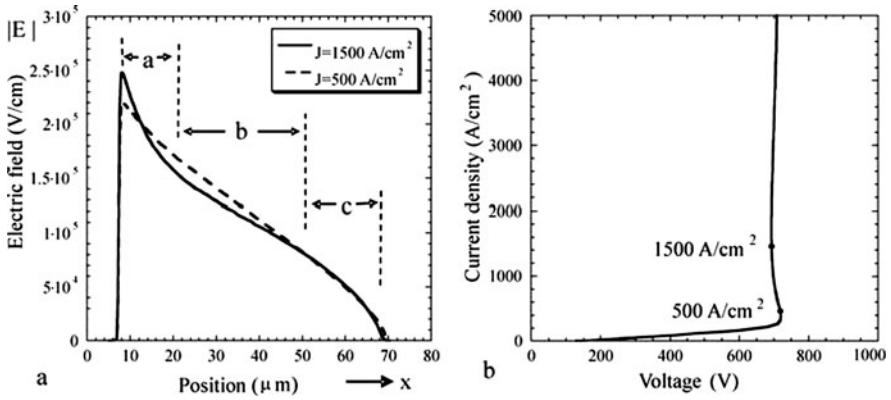


Fig. 12.15 Illustration for dynamic avalanche at an increased current density. Field shape (left-hand side), *I–V* characteristic (right-hand side). Reprinted from [Lut03] with permission from Elsevier

Close to the pn-junction – region a – the hole density is increased by the holes generated by avalanche. At the pn-junction holds

$$\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{q}{\varepsilon}\left(N_\mathrm{D} + p + p_\mathrm{av}\right) \tag{12.15}$$

With this, $\mathrm{d}E/\mathrm{d}x$ at the pn-junction has become very high. This is drawn for $j = 1500$ A/cm^2. Further away from the pn-junction, in region b, the electrons n_av generated by dynamic avalanche are flowing to the right-hand side. There are also some generated holes p_av, but their density is decreasing with distance to the pn-junction. It can be summarized for region b:

$$\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{q}{\varepsilon}\left(N_\mathrm{D} + p + p_\mathrm{av} - n_\mathrm{av}\right) \tag{12.16}$$

In region b a partial compensation of generated electrons and the flowing holes occurs, p_av is decreasing to the right-hand side, while n_av is increasing. Hence, the gradient of the electric field $\mathrm{d}E/\mathrm{d}x$ becomes flat. At the border to the plasma layer $E = 0$ must hold. Therefore in region c the gradient $\mathrm{d}E/\mathrm{dx}$ must increase again. The field in regions a to c will form a typical bowed shape.

The voltage correlates to the area under $E(\mathrm{x})$ which is slightly lower for the higher current density (see Fig. 12.15b). The *I–V* characteristic has a region of weakly negative differential resistance. Such a characteristic splits a homogenously distributed current to areas with lower current density and in filaments with high current density. The type of characteristic in Fig. 12.15b is described as S shaped in [Wak95], which can lead to the formation of stable current filaments.

Device simulation of such events shows a current density of 1000–2000 A/cm^2 in the filaments. Nevertheless, there are counteracting mechanisms, and destruction should be avoidable for the following reasons:

– The differential resistance is weakly negative, so that a further increase of current increases the voltage again. Therefore, the current density in the filaments is limited.
– The temperature inside the filament will increase leading to a reduction of impact ionization in the area of the filament, this counteracts the filamentation.
– The high local current density in a filament quickly removes the stored carriers locally from the plasma layer, counteracting the driving force for dynamic avalanche.

Therefore a state of rising, expiring, jumping, or moving filaments is to be expected, but a diode should still be capable to withstand this effect. Moving and jumping filaments are to be seen in two-dimensional device simulations [Nie05, Hei06]. It must be mentioned that two-dimensional simulations are limited, because filaments have a three-dimensional effect. However, such simulations are enormously time consuming and have not been possible up to now. In real devices, the edge of the active area is a strong inhomogenity, and it triggers the first filaments [Hei07]. For such a geometry including the edge of the active area, a two-dimensional simulation is capable of showing the main effects.

Most simulations in this field have been done under isothermal conditions that means without consideration of any temperature increase in filaments. However, the avalanche coefficients are strongly temperature dependent. The first temperature-dependent simulations of such problems showed an influence of temperature, filaments building up, and decreasing at different speeds, compared to isothermal simulations. However, the qualitative behavior was similar [Hei07].

Dynamic Avalanche of the Third Degree

Further increase of the stress by dynamic avalanche leads to the situation that a field at the nn^+-junction occurs while there is still dynamic avalanche at the pn-junction. The effects are shown in Fig. 12.16 in a simplified way. There is still remaining plasma, and the electric field between the plasma, and the pn-junction is bowed as in dynamic avalanche of the second degree. Between the plasma and the nn^+-junction a second electric field builds up, whose peak is at the nn^+-junction and whose gradient is inverted compared to the left space charge zone.

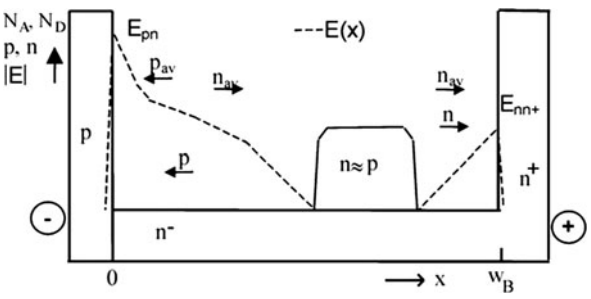


Fig. 12.16 Effects at strong dynamic avalanche. Fig adapted from [Lut09] © 2009 IEEE

This inverted gradient of the electric field can only occur, if the density of free electrons is so high that it overcompensates the positively charged donors of the background doping:

$$\left|\frac{\mathrm{d}E}{\mathrm{d}x}\right| = \frac{q}{\varepsilon}\,(n + n_{\mathrm{av}} - N_{\mathrm{D}}) \tag{12.17}$$

If n_{av} is increasing, the absolute value of dE/dx will increase, and finally impact ionization sets on at the nn$^+$-junction also. This is usually initiated locally in a region where a high current density filament at the pn-junction has formed. A double-sided dynamic avalanche is given at this position.

Figure 12.17 shows results of a simulation of a diode under such conditions [Hei08]. In Fig. 12.17a the current density is shown, between $y = 3600$ and 4400 µm a wide filament at the pn-junction has formed, at the nn$^+$-junction occurs a narrow high-current filament at $y = 4000$ µm. In Fig. 12.17b the electric field is shown for

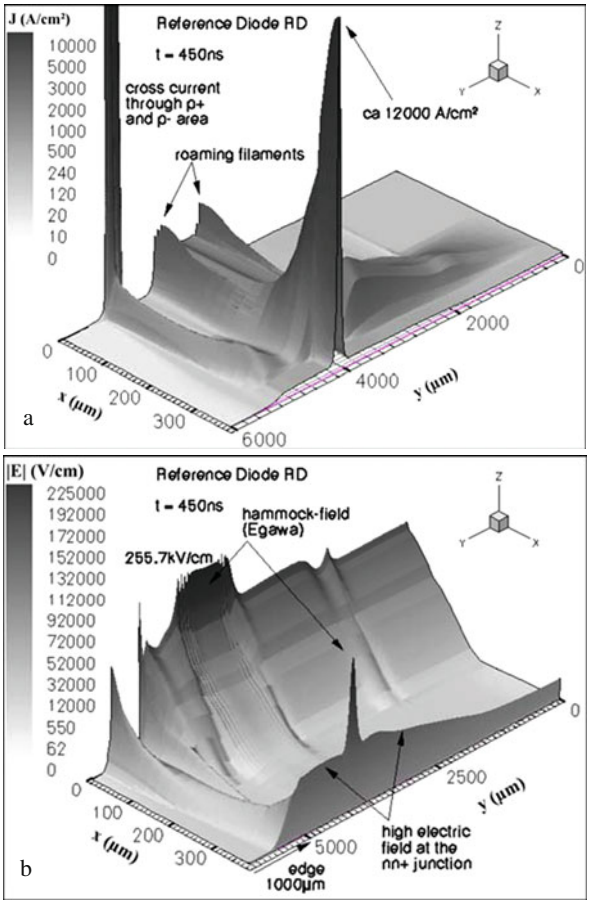


Fig. 12.17 Effects during strong dynamic avalanche: Current filaments and electric field in simulation of the turn-off of a 3.3 kV diode. Conditions $V_{\mathrm{bat}} = 1800$ V, $\mathrm{d}i/\mathrm{d}t = 1600$ A/µs, $L_{\mathrm{par}} = 1.125\,\mu H$. (**a**) Current density distribution at strong dynamic avalanche, current tube at the pn-junction, rising of a high current filament at the nn$^+$-junction at $x = 4000$ µm. (**b**) Electric field distribution, rising of an Egawa-type field at $x = 4000$ µm [Hei08]. Fig from [Lut09] © 2009 IEEE

the same instant. At the nn^+-junction a second electric field appears with a high field peak at $y = 4000\,\mu m$.

This hammock-shaped electric field has already occurred as an Egawa-type field [Ega66] in Sect. 12.3. There are also similarities to the field at the nn^+-junction at second breakdown in bipolar transistors. Applying Wachutka's model for the destruction limit of GTO thyristors [Wac91] to fast recovery diodes using the boundary conditions in [Ben67] leads to the conclusion, that dynamic impact ionization at the pn-junction should be stable, whereas impact ionization at the nn^+-junction is highly unstable. It is sufficient, that the ionization integral $\int_0^w \alpha\,(E(x))\,dx$ (see Eq. (3.67)) reaches a value of 0.3 to get to an instability mode. The current density in a filament will increase with a time constant in the range of some nanosecond [Dom03]. Impact ionization at the nn^+-junction is triggered by electrons and will occur at lower electric field strength because of the higher ionization rates of electrons. The rapid increase of the local current density is assumed here to be the cause of destruction.

In recent work, it has been shown by numerical simulation that the plasma-layer expands to the nn^+-junction in the vicinity of a cathode-side filament. This behavior differs from anode side filaments and results from the velocity saturation of the electrons and holes in the high-field region [Bab09]. The reduced depletion layer in the vicinity of the filament inhibits the lateral movement of the filament and, therefore, leads to strong local heating. This generates a destructive thermal filament.

The assumption that an Egawa-type field above a certain limit of 100 kV/cm at the nn^+-junction (Fig. 12.16, right-hand side) leads to the destruction of the diode could explain an experimentally found destruction limit. Figure 12.18 shows the

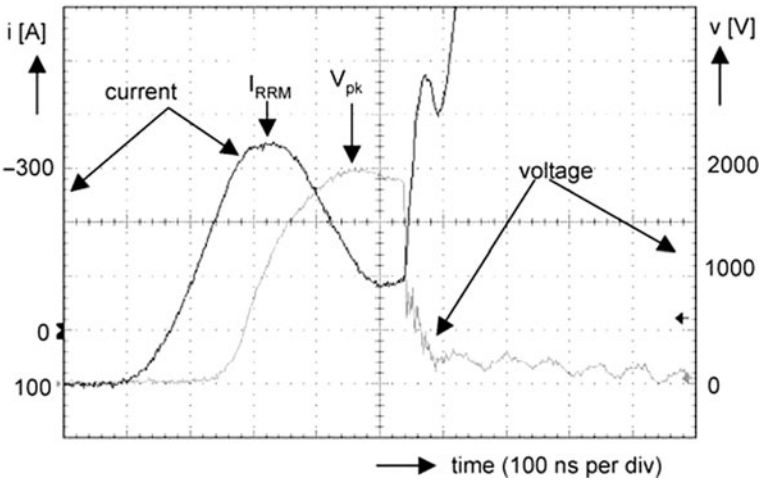


Fig. 12.18 Failure of a 3.3 kV diode under extremely strong dynamic avalanche conditions. Reprinted from [Lut03] with permission from Elsevier



Fig. 12.19 Failure picture of a diode destroyed by dynamic avalanche of the third degree

failure of a 3.3 kV-rated diode under the condition of very fast commutation di/dt. The reverse recovery current peak I_{RRM} of 360 A represents a reverse current density of 400 A/cm^2. The voltage across the diode is rising very fast, since in this experimental setup an additional capacitor of 22 nF was built-in between the gate and emitter of every switching IGBT. Already 200 ns after I_{RRM} the voltage has climbed up to 2000 V. The diode is destroyed shortly after the voltage peak.

If the employed IGBT withstands the short-circuit stress, which occurs after the destruction of the diode, and if it turns off the current successfully one can find a characteristic failure picture as shown in Fig. 12.19. At one point of the active area a small molten channel is found. In the case of dynamic avalanche of the third degree, cracks in angles of 60° were observed. The failure picture corresponds to the destruction of a 111-oriented silicon wafer by a point-shaped stress. This is a hint for a current filament of very high current density and very high temperature in a small area.

Such a failure limit was reproduced for devices of the same type of different production lots. Figure 12.20 shows the measurement results (black dots). The x-axis in

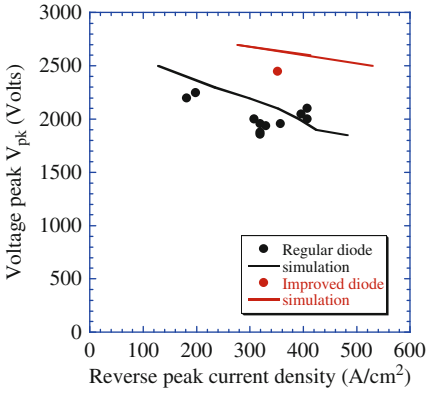


Fig. 12.20 Experimental failure limit and simulated failure limit. Reprinted from [Lut03] with permission from Elsevier

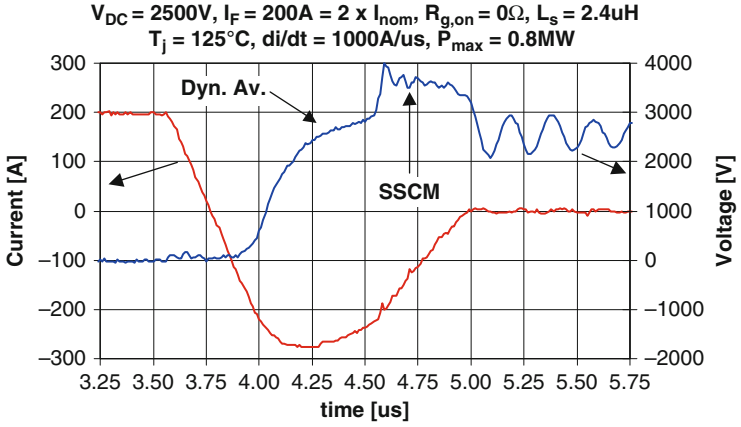


Fig. 12.21 3.3 kV diode with strong dynamic avalanche and SSCM mode [Rah04] © 2004 IEEE

Fig. 12.20 corresponds to the current density at reverse recovery current peak I_{RRM}, which can be adjusted by variation of di/dt. The y-axis shows the value of the voltage peak V_{pk} (see Fig. 12.18) which occurred before failure. It can be recognized that there is a limit for the voltage and this limit depends only weakly on the current density in the measured interval.

A device simulation with AVANT Medici resulted in the strait line shown in Fig. 12.20. As condition for failure it was taken that a second electric field builds up at the nn$^+$-junction and that a failure occurs if this field grows up to 100 kV/cm.

The simulation predicts that an Egawa-type field occurs at higher voltages if the width of the n$^-$-layer of the diode is increased (dotted line in Fig. 12.20). This was confirmed by an experiment for a diode with a 50 µm thicker middle layer. However, the experiments yield that the limits of failure are lower than the simulated limits in tendency. Nevertheless, Fig. 12.20 shows that with the selection of the necessary thickness, the ruggedness can be adjusted.

More recent results showed that freewheeling diodes even up to the range of 3300 V could achieve a very high capability to withstand dynamic avalanche [Rah04]. Figure 12.21 shows a measurement with very high stress on the diode by the high battery voltage $V_{\mathrm{bat}} = V_{\mathrm{DC}}$, high d$i$/d$t$, and especially a high inductance of 2.4 µH. Already shortly after the voltage increase, strong dynamic avalanche occurs, and above a voltage of 1500 V at $t = 4.1$ µs a flattening in the increasing voltage is observed, the reverse recovery current peak is widened and the voltage increases very slowly. After strong dynamic avalanche, at $t = 4.55$ µs the internal plasma is suddenly exhausted. The hole current, which before this instant is fed by the internal plasma, suddenly disappears, thus the reason for dynamic avalanche is no longer present. Now the voltage climbs up steeply, but it is limited by the diode itself. This limit at approx. 3700 V is close to the static avalanche breakdown voltage of the diode.

However, in the current waveform in Fig. 12.21, at this point in time occurs no snap off, but only a small dip. Thereafter follows a current generated in static avalanche. This mode was called "switching self-clamping mode" (SSCM) [Rah04], where the diode clamps the voltage peak.

In the SSCM mode holds [Hei05]:

$$\frac{\mathrm{d}i}{\mathrm{d}t} = \frac{V_{\mathrm{SSCM}} - V_{\mathrm{bat}}}{L_{\mathrm{par}}} \tag{12.18}$$

where V_{SSCM} is a voltage close below the static breakdown voltage V_{BD}. The capability of the diode to transit from the mode of dynamic avalanche into static avalanche is a high progress in ruggedness. According to [Rah04] the design of this diode observed the necessary measures to achieve a high ruggedness: The p^+-anode layer was highly doped. The edge of the active area is designed in a way that local current crowding is avoided. Especially, the nn^+-junction of this diode has a very shallow gradient. If at the nn^+-junction the doping increases slightly, N_{D} is increased, according to Eq. (12.17), and more electrons would be necessary for inversion of the gradient of the electric field, and the danger that an Egawa-type field can arise is lessened.

12.4.3 Diode Structures with High Dynamic Avalanche Capability

Increased doping at the n^+-side by a buffer reduces Egawa-type fields, since the increased density of positively charged donor ions compensates part of the arriving electrons. However, most efficient are structures which inject holes for compensation of avalanche-generated electrons.

One of these structures is the "Field Charge Extraction" (FCE) structure [Kop05]. A part of the cathode area contains a p^+-layer, as shown in Fig. 12.22, at the cathode side. At the anode side, a He^{++}-implantation was applied to reduce locally the lifetime and to adjust for soft recovery behavior, see Chap. 5. Soft recovery can be achieved with different methods. The effect of the FCE structure is at the cathode side: If a space charge builds up at the cathode side, the p^+-layer will inject holes. The injected holes compensate the electrons generated by dynamic avalanche.

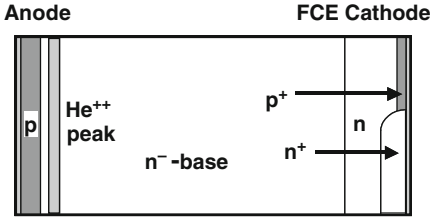


Fig. 12.22 Field Charge Extraction (FCE) diode, drawn after [Kop05] © 2005 IEEE

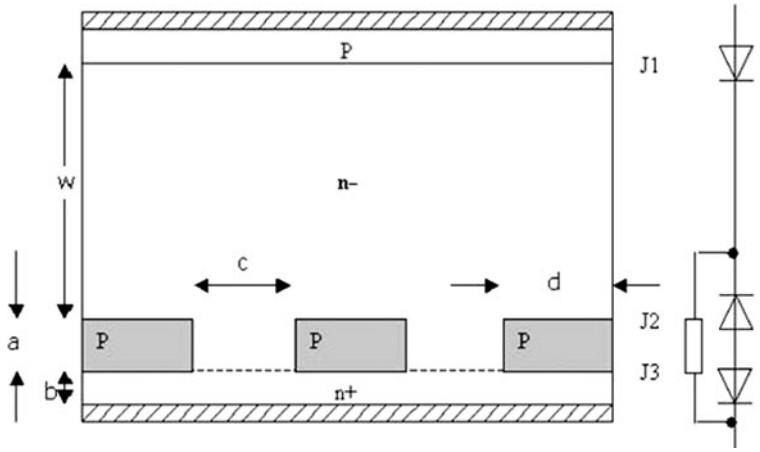


Fig. 12.23 Controlled injection of backside holes" (CIBH) diode [Chm06] © 2006 IEEE

The disadvantage of this structure is that part of the cathode area is lost for injection of carriers into the internal plasma. Therefore, the forward voltage drop of the diode will be increased.

The "Controlled Injection of Backside Holes" (CIBH) structure avoids this disadvantage. It contains floating p-layers in front of the cathode n^+-layer [Chm06]. The structure is shown in Fig. 12.23.

A continuous p-layer at the backside would realize a four-layer diode, this structure was discussed in [Mou88]. Such a four-layer diode acts in a similar way as a thyristor. It conducts in the forward direction after the cathode side pn-junction is overcome by breakover triggering. This leads to an additional voltage peak at turn-on of the diode and makes the structure unusable. Therefore in the CIBH diode this p-layer is interrupted by areas of the distance c, which form a resistor parallel to the additional junctions J_2 and J_3. The thickness of the p-layer between J_2 and J_3 is very low. Both sides of the junction J_3 are highly doped zones, which results in the onset of the avalanche breakdown at a small reverse bias like in an avalanche diode. Correspondingly the CIBH diode can be treated as a pin-diode with an integrated avalanche diode, which is connected with a parallel resistor. The distance c must be wide enough to avoid a deterioration of the turn-on behavior and it must be small enough to avoid high fields at the cathode side.

The effect of the suppression of electric fields is shown in Fig. 12.24. For comparison, Fig. 12.24a shows the electric field distribution during reverse recovery of a reference diode. The formation of the electric field peak at the nn^+ junction starts at $t = 400$ ns and leads to a critical Egawa-type field distribution after $t = 400$ ns, as described before.

Under the same switching condition the CIBH diode presents a completely different transient electric field distribution in the diode when compared to the reference diode, as illustrated in Fig. 12.24b. The situation at the backside is obviously improved. After the onset of the controlled avalanche at junction J_3, the voltage drop

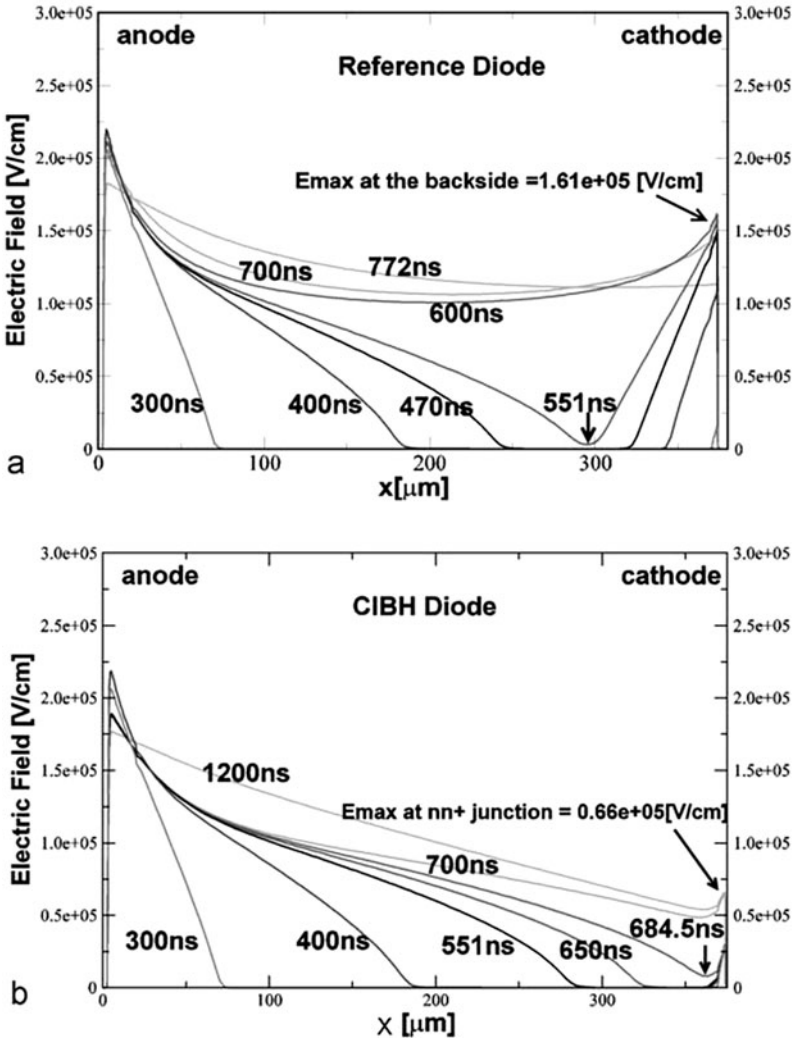


Fig. 12.24 Simulated electric field distribution of a reference diode (**a**) and of the CIBH diode (**b**) at very high stress in dynamic avalanche $T = 300\,\mathrm{K}$, $J_\mathrm{F} = 100\,\mathrm{A/cm^2}$, $V_\mathrm{bat} = 2500\,\mathrm{V}$, $\mathrm{d}i/\mathrm{d}t = 2000\,\mathrm{A}/\mu\mathrm{s}\,\mathrm{cm}^2$, $L_\mathrm{par} = 1.25\,\mu H$. Figure from [Chm06] © 2005 IEEE

at the backside is clamped at the breakdown voltage of the junction J_3. Avalanche at J_3 injects as much holes as necessary to compensate the electrons. The evolution of a second peak of electric field strength, which is correlated to a space charge region at the cathode side, is successfully suppressed.

Since Egawa-type fields can be effectively avoided, the CIBH diode has an extremely high ruggedness in dynamic avalanche. Figure 12.25 shows the turn-off of two 3.3.kV CIBH diodes under extreme stress. The maximal power density at turn-off is 2.5 MW/cm^2 – a factor of 10 above formerly assumed limits!

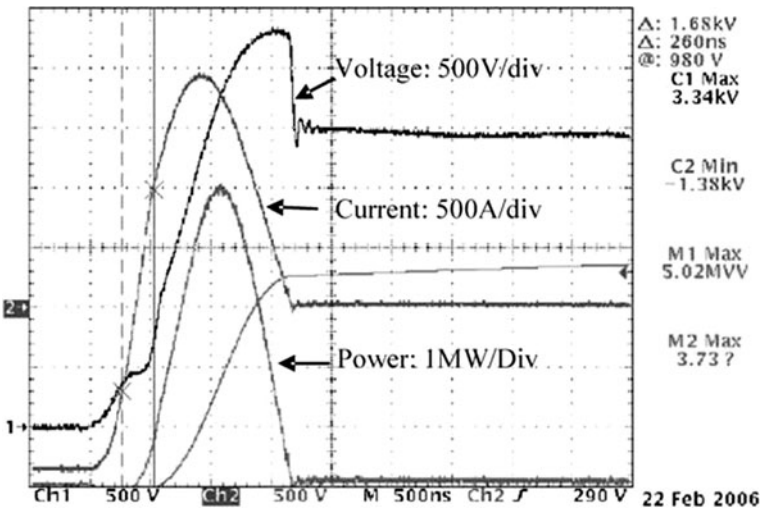


Fig. 12.25 Experimental ruggedness measurements of two CIBH diodes in a module at 5.5× nominal current and $T = 400\,\mathrm{K}$, $V_{\mathrm{bat}} = 2500\,\mathrm{V}$, $\mathrm{d}i/\mathrm{d}t = 6500\,\mathrm{A/\mu s}$, $L_{\mathrm{par}} = 0.75\,\mu H$, voltage (CH1), current (CH2), dissipated power (Math1). Figure from [Chm06] © 2005 IEEE

The CIBH diode has additional advantages in the turn-off behavior: during plasma removal, the plasma does not detach the cathode zone, but remains connected to the cathode, see Fig. 12.24b. In contrast to the usual process at reverse recovery as described in Fig. 5.26, the plasma in the CIBH diode is removed only from the anode side.

This effect occurs, since early during reverse recovery, junction J_3 goes into the avalanche mode and injects a current $j_{\mathrm{p,ava}}$. Then for the movement of the cathode-side plasma layer front holds [Bab08]

$$v_{\mathrm{R}} = \frac{j_{\mathrm{p}} - j_{\mathrm{p,ava}}}{q \cdot n} \tag{12.19}$$

The absolute value of v_{R} is reduced and if $j_{\mathrm{p,ava}} = j_{\mathrm{p}}$, v_{R} becomes 0. The plasma stays connected to the nn$^+$-junction. In Eq. (5.94) v_{R} is equal to 0 and we obtain $w_{\mathrm{x}} = w_{\mathrm{B}}$. Therefore the whole base width w_{B} is supporting the voltage V_{sn} and in Eq. (5.95) $w_x = w_{\mathrm{B}}$ holds.

If now the space charge region reaches w_{B} – this is the case at turn-off under conditions of low forward current, high parasitic inductance, and high voltage, where usually freewheeling diodes show snappy recovery – the p-layers inject additional holes. If the applied voltage is further increased, the reverse recovery is further improved. This was introduced as DSDM (dynamic self-damping mode) [Fel08].

The forward voltage drop of the CIBH diode is not distinctly increased compared to another diode of comparable thickness and stored charge, since during forward conduction we have a parallel connection of a triggered thyristor and a diode. A possible drawback could be a loss in the static breakdown voltage V_{BD} because of

the hole injection into the electric field. By adjusting the process parameters even at high p-doses and high p area ratios the loss in the static breakdown voltage can be avoided while the softness is improved. The loss of breakdown voltage is in maximum 7% for diodes with a high implantation dose [Fel08].

12.4.4 Dynamic Avalanche: Further Tasks

The CIBH diode shows that power devices can be designed in a way that they can withstand high stress in dynamic avalanche. A review on dynamic avalanche in high-voltage devices is given in [Lut09]. However, manufacturers still restrict the allowed reverse current I_{RRM} with safe-operation area (SOA) diagrams. This SOA diagrams are usual for devices with blocking voltages of 3.3 kV and above and have a shape similar to Fig. 12.12. Such diagrams forbid the application of devices in dynamic avalanche.

We also conclude from Fig. 12.12: the higher the rated voltage of the device, the more the allowed reverse current density must be restricted if one wants to avoid dynamic avalanche. On the other hand, the stored charge Q_{RR} in a device increases with the base width of a device at a power of two – see Chap. 5, Eq. (5.64). With increased Q_{RR} also I_{RRM} is increasing. To keep I_{RRM} low, the current slope di/dt must be kept small. However, with a small di/dt during turn-on of a transistor, the turn-on losses in the transistor increase. Therefore, the required restriction of I_{RRM} restricts the possibility to reduce the switching losses in the application.

It was found that the higher the rated voltage, the lower the current densities at which formation of filaments and current tubes occur [Nie04]. Therefore, in spite of the progress, further research is necessary to understand the behavior in power devices at the border of the safe operation area. Similar effects of negative differential resistance and moving filaments have been found in ESD protection devices [Pog03]. Meanwhile one can understand some experimental results and attempts for improved designs have been successful, but there still exist open questions.

12.5 Exceeding the Maximum Turn-Off Current of GTOs

During turn-off of a GTO thyristor, the current below the emitter finger is extracted from the edge toward the middle. This was shown in the paragraph on GTO thyristors, see Fig. 8.16. Finally a narrow current-conducting area remains in the middle of the emitter finger, representing a current filament before the anode current decays. Even if high accuracy of the fabrication process technology is given, not all emitter fingers of the GTO thyristor are ideally identical, and even in a single emitter finger, there will be a position which will be the last for current conduction. If the maximum turn-off current is exceeded, a molten zone is found at this point. After chemical removal of the metallization and etching in concentrated potash KOH, which solves polysilicon faster than mono-crystalline silicon, a narrow hole can be found as shown in Fig. 12.26.



Fig. 12.26 GTO thyristor failed by exceeding the maximum turn-off current. Molten channel, created by a current filament in the middle of the cathode finger

The dark area in Fig. 12.26 shows the position of the final filament or current tube, while the molten channel reaches down to the backside of the device.

A failure picture as in Fig. 12.26 can be caused by exceeding the maximum turn-off current capability of the device. However, it can also occur because of a failure of an element in the RCD snubber (see Fig. 8.18), since during turn-off of a GTO thyristor the voltage slope dv/dt must be limited.

12.6 Short-Circuit and Over-Current in IGBTs

12.6.1 Short-Circuit Types I, II, and III

Three types of short circuit have to be distinguished for the IGBT [Eck94, Eck95, Let95]:

Short circuit I is a direct turn-on of the IGBT to a short circuit. The course of the voltage V_C, the collector current I_C, and the gate voltage V_G is shown in Fig. 12.27. Before turn-on, the voltage V_C is high, while at the gate a negative voltage is applied. After turn-on into the short circuit, the current increases to more than 6 kA, this is the value of the saturation current. This value is specified in some data sheets as I_{SC} and correspondents to the saturation current at $V_G = 15$ V. The IGBT is able to withstand the simultaneous load of a high current and a high voltage during the short-circuit pulse for some time. It must be turned off within a defined time, typically specified to 10 µs or lower, to ensure safe operation and to avoid failures by overheating.

In Fig. 12.27 I_{SC} decreases with time due to self-heating of the device. During turn-off of the short-circuit current, an inductive voltage peak is generated. In the figure this voltage peak amounts to approx. 2000 V. The condition for surviving of a short-circuit pulse is that the peak voltage must remain lower than the rated voltage within the specified safe operation area. To ensure this, short circuit must be turned off with a limited di/dt. Usually, for turn-off of the short circuit the driver uses a higher gate resistor to limit the di/dt. This soft turn-off is visible in Fig. 12.27.

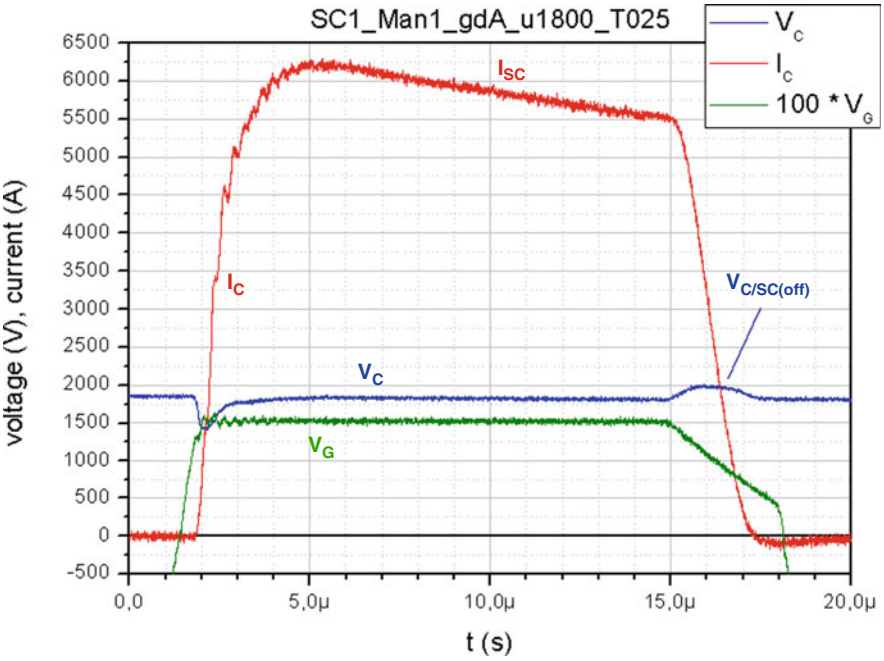


Fig. 12.27 Short circuit I. Figure from [Lut09b] © 2009 EPE

Short circuit II is the occurrence of a short circuit during the conducting mode of the IGBT [Eck94, Let95, Ohi02]. The occurrence is shown in Fig. 12.28 for the same IGBT as in Fig. 12.27. The IGBT is carrying the load current I_{Load} and the voltage drop is V_{CEsat}. As soon as the short circuit has occurred, the collector current will increase very steeply. The di/dt is determined by the DC-link voltage V_{bat} and the inductance of the short-circuit loop. During the time interval I, the IGBT is desaturated. The consequently high dv/dt of the collector–emitter voltage will induce a displacement current through the gate–collector capacitance, which increases the gate–emitter voltage. As can be seen in Fig. 12.28, V_{G} is dynamically increased up to 20 V due to this effect. This increase of V_{G} in turn causes a high short-circuit current peak $I_{\mathrm{C/SC(on)}}$, which in this case reaches a value of 14 kA.

Since the gate–collector capacitance is high at low V_{C}, the effect of the displacement current will be most significant at low voltage. The increased gate voltage during desaturation leads to a negative gate current back to the driver. The onset of a negative gate current together with the interconnection parasitics and IGBT input capacitances are supposed to cause the oscillations in V_{G} showed in Figs. 12.28 and 12.30.

After having completed the desaturation phase, the short-circuit current will drop to its static value I_{SC} (time interval II). Due to the current fall with negative di/dt, a voltage $V_{\mathrm{C/SC(on)}}$ will be induced over the parasitic inductances, which becomes visible as an voltage peak on the IGBT:

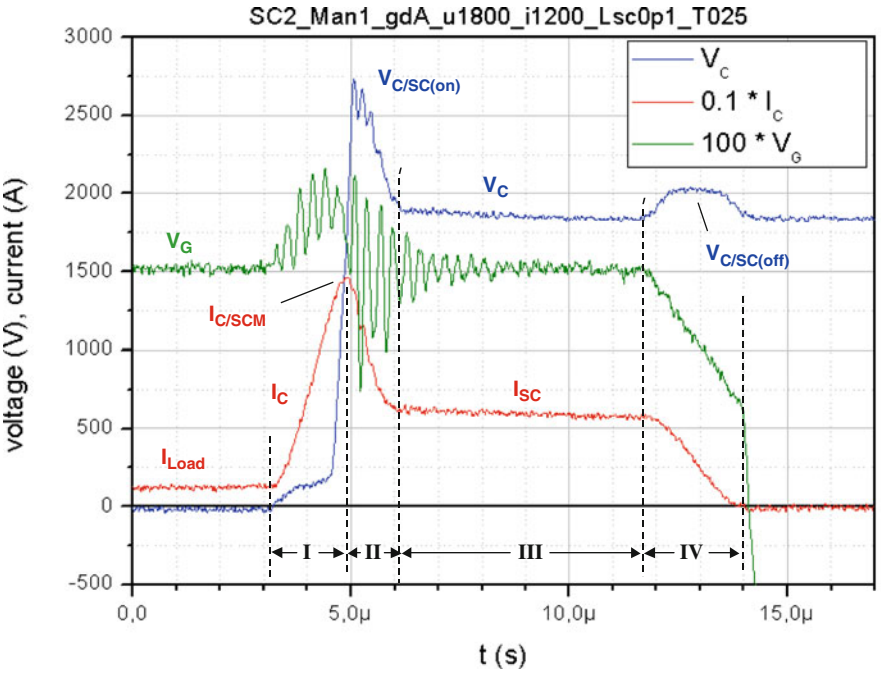


Fig. 12.28 Short circuit II. Figure from [Lut09b] © 2009 EPE

$$V_{\mathrm{C/SC(on)}} = V_{\mathrm{bat}} + L_{\mathrm{SC}} \cdot \left| \frac{\mathrm{d}i_{\mathrm{C}}}{\mathrm{d}t} \right|_{\mathrm{max}} \tag{12.20}$$

The stationary short-circuit phase (time interval III) is followed by the turn-off of the short-circuit current. Due to the negative di/dt, the commutation circuit inductance will again induce an overvoltage $V_{\mathrm{C/SC(off)}}$ across the IGBT (time interval IV). Once again it must be ensured that this voltage peak stays below the rated voltage, respectively, within the specified safe operation area. To limit $I_{\mathrm{C/SCM}}$ and to keep the gate–emitter voltage within the permissible limits, V_{G} has to be clamped.

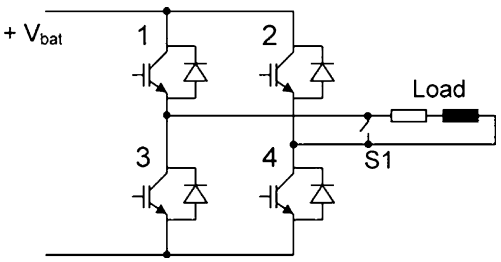


Fig. 12.29 Explanation of an SC III on example of a single phase inverter

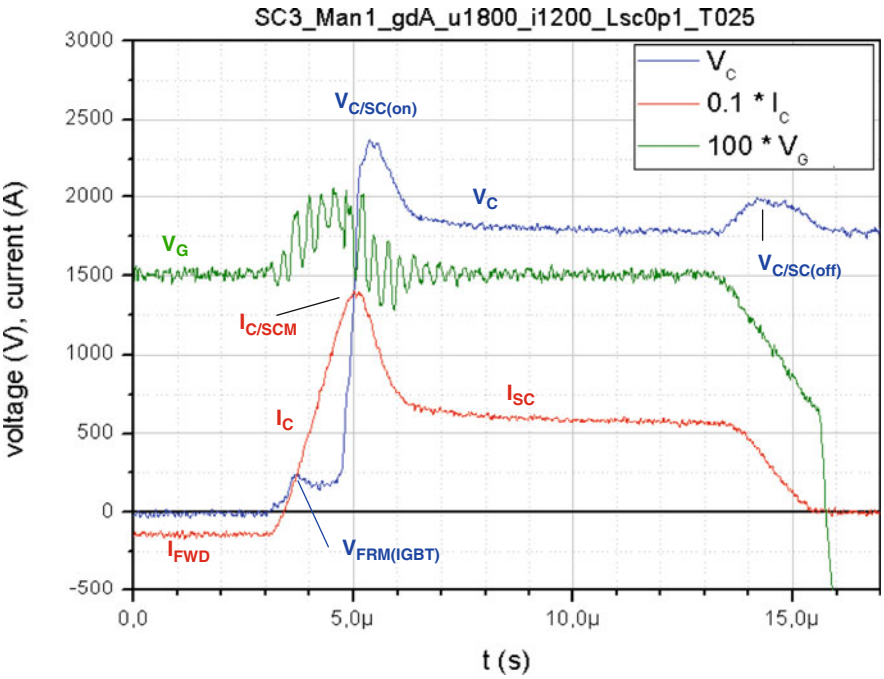


Fig. 12.30 Short circuit III. Figure from [Lut09b] © 2009 EPE

It should be noted that short circuit II is a harder condition than short circuit I. While $V_{C/SC(off)}$ can be limited by a soft turn-off function of the driver, this does not hold for $V_{C/SC(on)}$, which can easily overshoot the rated voltage of the device.

Short circuit III is the occurrence of a short circuit across the load during the conducting mode of the freewheeling diode. SC III can occur in all typical IGBT applications, since there is always an interval where the freewheeling diode is conducting. In motor drive applications, the duty cycle for the IGBT is higher than for the freewheeling diode, and SC II is more probable. However, if, e.g. a train drives down from a mountain, the motor is used as generator and energy is transferred back from the train to the grid. The duty cycle for the freewheeling diode is higher for the diode as for the IGBT in this operation mode and therefore SC III is more probable to occur than SC II.

The occurrence of a SC III event is explained using Fig. 12.29, in which a single-phase inverter is shown.

Inverters with pulse width modulation usually are driven with complementary signals. As starting point we assume that IGBT 1 and IGBT 4 are on and the current flows through IGBT 1, the load, and IGBT 4. As next, the IGBTs 1 and 4 are turned off, the gate signal of IGBTs 2 and 3 is set to "on" after a short dead time. Because the inductance of the load determines the current direction, the current will in this

case flow through diodes 2 and 3 back to the voltage source. If now a short circuit occurs across the load – symbolized by closing the switch S1 – diode 2 and 3 will be commutated with a high di/dt and IGBTs 2 and 3, which at this time interval have a gate signal which is already set to "on," will be exposed to a short circuit. The current is thereby rapidly commutated from the diode to the IGBT, and the IGBT is turned on passively into the short circuit. During the short-circuit event, a reverse recovery process occurs at the freewheeling diode; however, the voltage course is determined by the IGBT and the occurring high dv/dt while the IGBT transits to its desaturation mode.

Before the short circuit, a positive $V_G = 15$ V is already applied at the IGBT; however, its current is zero because the inverse diode is conducting at this stage in the PWM pattern. When the short circuit occurs, the current changes its sign and the IGBT is turned on in a passive, diode-like manner. At usual turn-on, the voltage across the IGBT is high before turn-on. In the now given case, the voltage is low. Therefore a forward recovery peak V_{FRM} at the IGBT occurs. This is related to the forward recovery of a diode. It was observed in [Pen98] in a specially designed zero-voltage switching circuit with relatively low-voltage peaks. This forward recovery peak V_{FRM} can get very high, depending on di/dt. Several hundred volts can occur with wide-base high-voltage IGBTs. Forward recovery voltage peaks at IGBTs may be higher than at diodes [Bab09]. However, the measurement in Fig. 12.30 is at the outer terminals and the parasitic module inductance contributes significantly. Additionally, the diode parallel to the IGBT is in the reverse recovery process, the measured current at this instant contributes to diode and IGBT current and cannot be resolved in the used setup.

The dynamic short-circuit peak current $I_{C/SCM}$ amounts to 14 kA, this is almost the same value as in SC II. $I_{C/SCM}$ in SC III was found to be similar to $I_{C/SCM}$ in SC II for IGBTs from different manufacturers [Lut09b].

The occurrence of V_{FRM} is not supposed to be an extraordinary stress for the IGBT, since it occurs in the IGBT forward direction, where a high blocking capability is specified. The main additional stress in SC III for the module is the reverse recovery of the freewheeling diode during short circuit. The two-step voltage slope at reverse recovery of the diode, in which the second step may occur with an extraordinary high dv/dt, is a special stress event for the FWD. Usually, IGBTs fail in short circuit. However, in one of the SC III experiments, a diode failure was observed while the IGBT withstood [Lut09b].

A typical picture of a 3.3-kV IGBT, destroyed by short circuit is shown in Fig. 12.31. The large-area burned emitter regions are typical. A picture of a destroyed IGBT similar to Fig. 12.31 gives the specialist a hint that short circuit is probably the failure reason. The burned-off emitter regions are typically found for SC I as well as for SCII and SC III failures.



Fig. 12.31 IGBT die (3.3 kV) destroyed by short circuit

12.6.2 Thermal and Electrical Stress in Short Circuit

From its basic function, the IGBT has a short-circuit capability that limits the current. I_{SC} is the current in the active region of the IGBT forward *I*–*V* characteristic and can be approximated similar to Eq. (10.3):

$$I_{SC} = \frac{1}{1-\alpha_{pnp}} \cdot \frac{\kappa}{2} \cdot (V_G - V_T)^2 \,\big|_{V=V_{bat}} \tag{12.21}$$

An electric field exists in the device (SC I) or is built up at desaturation (SC II, SC III). In short circuit II, there might be remaining plasma from the saturation mode before short circuit. The electric field is built-up quickly and the remaining plasma is extracted in a short time. Figure 12.32 shows the process in an NPT-IGBT in the short-circuit mode. An electric field has built up at the blocking pn-junction, which takes over the applied voltage V_{bat}. On the emitter side, the left-hand side in Fig. 12.32, the n-channel is conducting. Electrons are flowing into the space charge region, holes are injected from the p-emitter. In the electric field, the carriers flow with their drift velocities given in Chap. 2 with Eq. (2.38). The current density j_{SC}

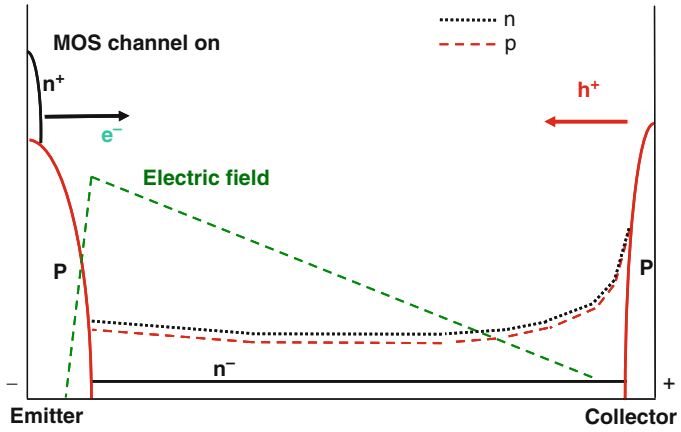


Fig. 12.32 Process in an NPT-IGBT during short circuit

is composed of $j_n + j_p$, where

$$\begin{aligned} j_n &= q \cdot n \cdot v_n \\ j_p &= q \cdot p \cdot v_p \end{aligned} \tag{12.22}$$

Since v_n, v_p are now much higher than in a plasma layer, the total amount of electrons and holes is much lower than at forward conduction where *n,p* are in the range of 10^{16} cm^{-3}. Sometimes $v_{sat(n,p)}$ is used for calculation of the carrier densities; however, especially for holes the drift velocity is often not saturated at the given fields. *n,p* are typically in the range of several 10^{14} cm^{-3} and clearly above the background doping N_D. The mobile carriers therefore lead to a feedback to the electric field and it holds

$$\frac{dE}{dx} = \frac{q}{\varepsilon}\left(N_D - \frac{j_n}{q \cdot v_n} + \frac{j_p}{q \cdot v_p}\right) \tag{12.23}$$

where the terms in the bracket form an effective doping $N_{eff} = N_D - n + p$. The detailed feedback depends on the used IGBT technology and the conditions. In PT-IGBTs with the applied high p-emitter efficiency, typically the term caused by the hole current is dominating, and N_{eff} will be increased. In NPT-IGBTs, the electron density *n* is above the hole density and the reduced d*E*/d*x* leads to a wider extension of the space charge into the n$^-$-layer, reducing the electric field at the pn-junction [Las92]. In IGBTs with buffer layer, even $n > N_D + p$ may occur, the gradient of the field will be inverted and the field peak shifts to the collector side.

The active region is a stable condition for a transistor; the deposited energy in the case of a SC I is

$$E = V_{bat} \cdot I_{SC} \cdot t_{SC}$$

and due to the simultaneous high V_{bat} and I_{SC}, the temperature increases fast. If the short circuit is turned off within the requested time interval – 10 µs for former IGBT generations, 7 µs for new generations – the IGBT can in most cases withstand the thermal stress occurring during the short-circuit mode.

The holes are flowing through the p-well close to the emitter layer; this was shown in Fig. 10.14. The voltage drop in this path must be well below the built-in voltage V_{bi} of the junction. On the one hand, the density of carriers in short circuit is much lower than at rated current, where the IGBT is flooded with plasma. On the other hand, during short circuit very high temperatures occur. A high temperature decreases V_{bi}, therefore the danger of latch-up increases with increasing temperature. Nevertheless, there was much progress in manufacturing highly doped p^+-wells, therefore latch-up is usually no more a limit for the short-circuit capability of latest IGBT generations.

Equation (12.21) has given the parameters that determine I_{SC}. α_{pnp} is adjusted to be low in modern IGBTs by a low emitter efficiency at the collector side. It is typically in the range of 0.33 – 0.4. The other decisive factor in Eq. (12.21) is the channel parameter κ which gives the conductivity of the MOS channel. Since a comparatively high cell distance is of advantage in modern IGBTs – see Sect. 10.6 – the channel parameter κ is kept moderate. Therefore, in modern IGBTs a low I_{SC} can be achieved despite a high plasma density at the emitter side during usual forward conduction. As an example, Fig. 12.33 shows a comparison of a modern and conventional IGBT [Mor07]. For the modern IGBT (HiGT), the saturation current is not increased, despite the lower V_C at the typical operation condition (denoted V_{CE}(sat) in Fig. 12.33). The I_{SC} of the 50-A rated IGBT is approximately 175 A. Since the thermal short-circuit capability depends on the deposited energy in short

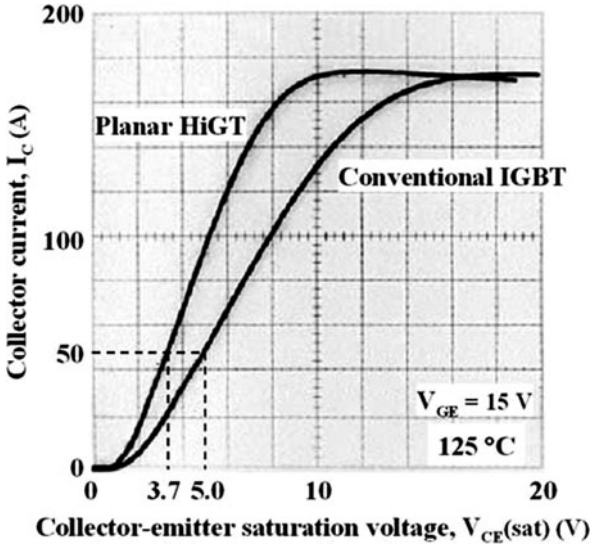


Fig. 12.33 Measured forward characteristic of a 3.3-kV conventional IGBT and planar HiGT (modern type). Figure from [Mor07] © 2007 IEEE

circuit, which is given by $I_{SC} \cdot V_{bat} \cdot t_{sc}$, the modern IGBT has the comparable short-circuit capability as the conventional one.

In several data sheets of 1200-V IGBTs, I_{SC} is even lower for the new IGBT generations. I_{SC} is specified to approx. 4 times the rated current, compared to 5–6 times the rated current for the older generations. In combination with a very high doping of the p-well (Fig. 10.14), leading to a very low resistance R_S (see Fig. 10.2b), very high short circuit ruggedness is achieved in modern IGBTs. However, in future IGBT generations with further reduced device area and device volume, the allowed time t_{sc} before turn-off of the short-circuit pulse will be limited from 10 µs to a lower value of 5 – 7 µs. This is possible in the application with meanwhile available fast-reacting improved gate drive units.

Thermal Limits for Medium-Voltage IGBTs

In medium-voltage IGBTs, the short-circuit capability is limited by temperature. The destruction of a 600-V IGBT by short circuit I is shown in Fig. 12.34. The applied stress in this case is beyond the short-circuit capability of the device. In Fig. 12.34a, the short circuit occurs at a battery voltage of 540 V during a time of approx. 60 µs, until the device is destroyed. In Fig. 12.34b the short-circuit current is turned off after approx. 40 µs. During the short-circuit pulse the device has been heated to a level that a high leakage current is flowing after the turn-off event. At such high temperatures additional charge carriers are generated thermally, see Eq. (2.6) and the leakage current increases further. After approx. 100 µs or more, the device is destroyed by overheat caused by the high leakage current.

Long-time tests with repetition of short-circuit events led to the conclusion that short circuit can be repeated up to 10,000 times without destruction of the device [Sai04]. This holds true as long as the deposited energy stays smaller than a certain critical energy E_C. A summary of the repeated short-circuit tests of a 600-V IGBT is shown in Fig. 12.35. A defined limit for the dissipated energy was found

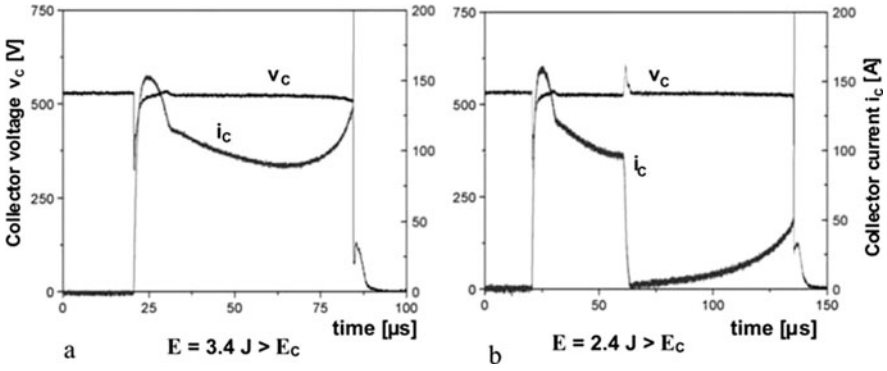


Fig. 12.34 600 V IGBTs in short circuit at $V_{bat} = 540\,V$, $T = 125°C$. Course of current and voltage at destruction of IGBTs at a deposited energy beyond the critical energy E_C. Figure from [Sai04]

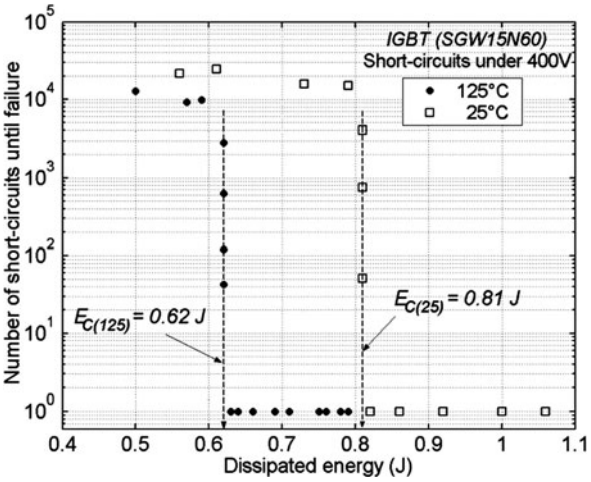


Fig. 12.35 IGBT Ruggedness at repeated short-circuit events. 15-A 600-V IGBT, $V_{bat} = 405$ V. Figure from [Lef05] © 2005 IEEE

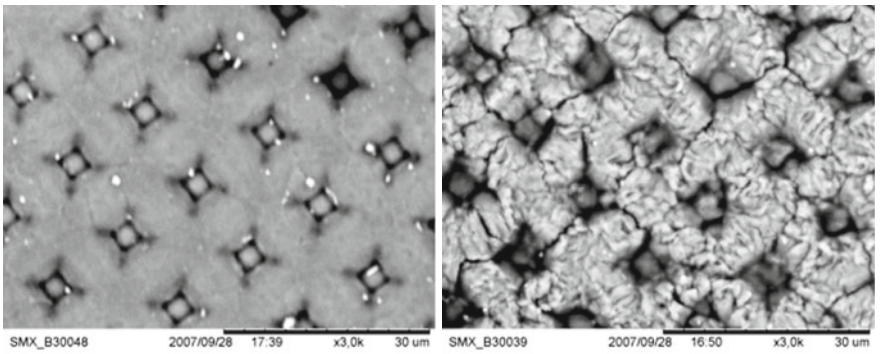
and this critical energy E_C is lower for the $T = 125°C$ compared to $T = 25°C$. Above the limit E_C, IGBTs are destroyed after one event by overheating. Since heat transport out of the device is small for the requested short time of the short-circuit load, the temperature increase can be calculated according to the thermal capacity of the device, compare Eq. (11.8), treating the deposited energy E as thermal energy Q_{th} by

$$\Delta T_{SC} = \frac{E}{C_{th}} = \frac{V_{bat} \cdot I_{SC} \cdot t_{SC}}{c \cdot \rho \cdot d \cdot A} \tag{12.24}$$

The evaluation of different IGBTs and the following calculation of the temperature increase for the volume of the semiconductor in which the electric field occurs resulted in temperatures in the range of 600°C [Sai04]. For IGBTs with different thicknesses, a similar final temperature was also found.

The results in Fig. 12.35 show that the failures in short circuit are purely thermal for the investigated 600-V IGBTs. Additionally, special attention was taken to find the ageing mechanisms of IGBTs which fail below E_C, but after a large number of short-circuit pulses. No trend in leakage current and threshold voltage was found. However, it was observed that the forward voltage drop V_C increases and the short-circuit current I_{SC} decreases with the number of cycles. Failure analyses showed the increase of the resistivity of the Al metallization layer, a strong degradation of the die metallization by Al reconstruction after approx. 10,000 cycles (Fig 12.36), and also a strong degradation of the bond wire attach [Lef08].

The degradation of metallization may lead to inhomogeneous current distributions and finally local overheating. The observed aging mechanisms have similarities to power cycling, where also the effects of aging of the metallization layer and bond wire lift off are observed [Lut08], see Sect. 11.6.



a b

Fig. 12.36 Reconstruction of Al metallization at repeated short circuits: (a) before test (b) after 24,600 short-circuit cycles. Figure from [Lef08]

Since the main destructive effects are due to temperature, the critical energy E_C, which the device can take up as heat, depends on its thermal capacity. Modern IGBTs are designed with narrow widths w_B for the low-doped n-base layers to reduce the overall losses. Additionally the reduced voltage drop V_C during forward conduction allows higher rated currents for a given device area. Therefore the area as well as the thickness of the IGBT die is reduced and the thermal capacity is decreased correspondingly.

An example of the manufacturer Infineon for the die area of different 1200-V 75-A IGBT generations is shown in Fig. 12.37. The thickness of the dies for different IGBT generations and three different voltage ratings is shown in Fig. 12.38. For

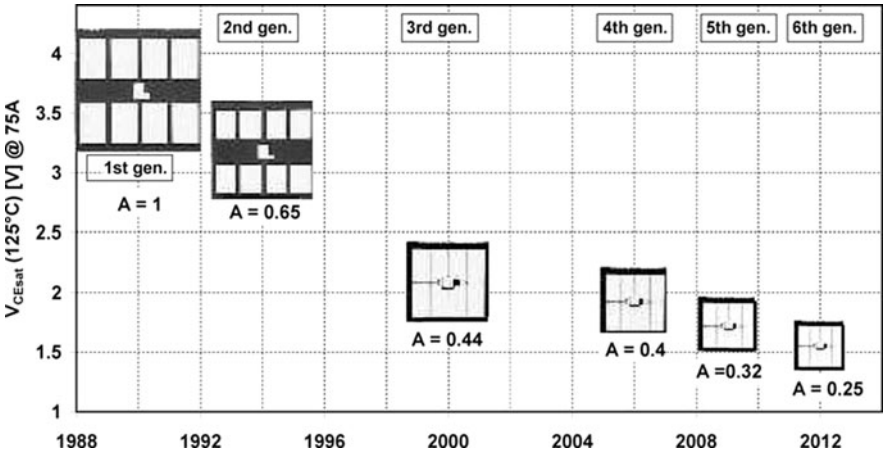


Fig. 12.37 Die area and forward voltage drop for different IGBT generations, on example of the manufacturer Infineon. Figure according to T. Laska, Infineon

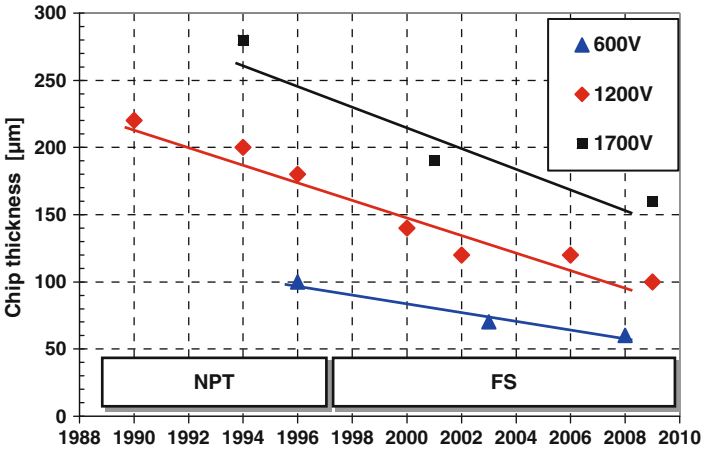


Fig. 12.38 Device thickness for different IGBT generations of the manufacturer Infineon. Figure from Infineon

the 1200-V 75-A IGBT chip, the area is reduced down to 44% and the thickness to 55% compared to the first IGBT generation from 1990. According to Eq. (11.8) this is a reduction of the thermal capacity down to 24%.

With this drastic reduction of the thermal capacity, also the thermal energy, which can be deposited in a device, is reduced in the same way. To maintain ruggedness during short circuit, I_{SC} has to be kept low.

Current Filamentation as Limit for the Short-Circuit Capability of High-Voltage IGBTs

While for 600-V and 1200-V IGBTs the short-circuit failure limit is mainly due to thermal reasons and aging [Lef08], this does not apply for high-voltage IGBTs. Due to the larger thickness of the device and due to the reduced current density, the temperature increase calculated with Eq. (11.24) is typically smaller. Especially, when short-circuit limits are determined, failures are not observed after a long t_{sc}, or at short-circuit turn-off, or after the turn-off have been completed, as shown in Fig. 12.34. They occur typically during the stationary phase in SC I. An example for such a measurement is shown in Fig. 12.39 [Kop09].

Figure 12.40 shows a summary of short-circuit failures for the 6.5 V IGBT as function of the applied DC-link voltage V_{bat}. At around $V_{bat} = 2000$ V the capability reaches a minimum and starts increasing again for higher V_{bat}. Note, the ruggedness is higher at 4500 V than at 1000 V!

A minimum for the short-circuit capability is found between 1500 and 2500 V, similar minima were found for a 4.5 kV rated IGBT between 1200 and 1800 V, and at about 1000 V for a 3.3 kV rated IGBT. This special voltage dependency needs new explanations.

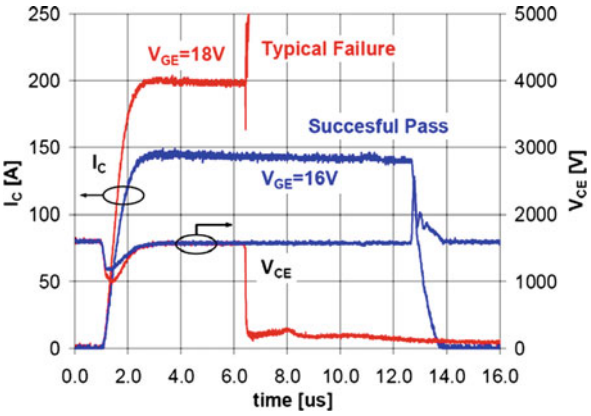


Fig. 12.39 Typical short-circuit pass and fail waveforms, SC I measurement of a 6.5 kV IGBT. Figure from [Kop09] © 2009 IEEE

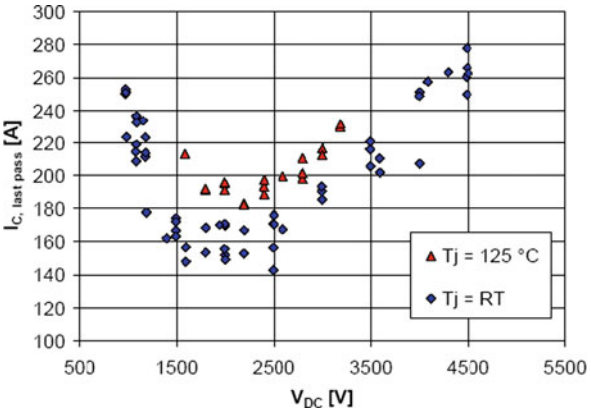


Fig. 12.40 Short-circuit capability for a 6.5 kV IGBT in dependency on the DC-link voltage V_{bat} and starting junction temperature. Figure from [Kop09] © 2009 IEEE

In these high-voltage IGBTs with a buffer layer at the collector side, a redistribution of the electric field to a field peak at the collector side under short-circuit stress was found [Kop08, Kop09], which results from a high electron density $n > N_D + p$, see Eq. (12.23). However, these field peaks at the nn^+-junction are moderate and the increasing ruggedness with increasing voltage excludes second breakdown and Egawa-type failure mechanisms. Further, it was found that an increased emitter efficiency of the collector-side p-emitter improves the ruggedness [Kop08, Kop09]. The increased α_{pnp} is, for the given low α_{pnp}, of low effect to I_{SC}. This experimental result also excludes latch-up as failure root cause. In [Kop09], the formation of filaments is suggested as failure mechanism. These filaments, found in numerical simulation, result from the different drift velocities of electrons and holes. Especially at medium voltage the electric fields are moderate and the drift velocity for the holes is far below that of electrons, see Fig. 2.14. Filaments with high local current density lead to IGBT destruction.

12.6.3 Turn-Off of Over-Current and Dynamic Avalanche

The turn-off of over-current is a very critical operation point of an IGBT. In difference to short circuit, the turn-off process is executed while the device is highly flooded with free carriers. The first IGBT generations for 3.3 kV restricted the turn-off capability of over-current to two times the rated current [Hie97]. A higher current was not allowed to be turned off. If a higher current occurs in these IGBTs, the driver has to wait until the IGBT is in the mode of saturation current, while the voltage increases and the IGBT then is turned off in the short-circuit mode. Note that in the short-circuit mode 5 – 6 times of the rated current are switched off without problems.

During turn-off of an over-current, the channel is first turned off. Hence, the electron current flowing via the channel becomes extinct. The total current must flow for a short instant completely as hole current. Figure 12.41 shows the process, for example, of an NPT-IGBT.

The hole current, fed by the remaining plasma, flows across the n^--layer, in which an electric field has build up. The density of free holes adds to the background doping. The gradient of the electric field gets steeper, as already treated with Eqs. (12.10) to (12.12). The blocking capability is therefore reduced as given in Eq. (12.13). Dynamic avalanche now generates electron–hole pairs in the region close to the blocking pn-junction. The holes are flowing to the left side in Fig. 12.41, while the electrons flow to the right side. The hole current, increased by dynamic avalanche, must flow through the p-well and via the resistor R_S (Fig. 10.2b). In this operation mode, the hole density is highest and the danger of turn-on of the parasitic npn-transistor and latching of the IGBT is greatest. If R_S is low enough, the IGBT will successfully withstand such conditions.

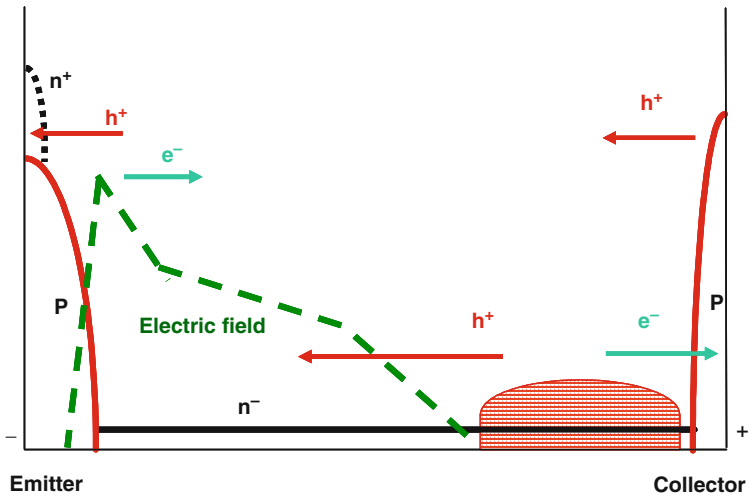


Fig. 12.41 NPT-IGBT at turn-off of an over-current and occurrence of dynamic avalanche

The electrons generated close to the pn-junction are flowing to the right-hand side and they compensate the hole current. An electric field will rise as drawn in Fig. 12.41 for a strong dynamic avalanche mode. This field has close similarity to the S-shaped field in dynamic avalanche of the second degree and the arguments given in context of Eqs. (12.15) and (12.16) hold, respectively.

An electric field shaped as in Fig. 12.41 leads to a weakly negative differential resistance in the *I*–*V* characteristics and an investigation of the effect using device simulation [Ros02] showed the occurrence of current tubes, which arise in certain regions and which jump to neighboring cells. This process also is similar to dynamic avalanche of the second degree.

However, a fundamental difference to a diode is that on the right side of the plasma a p-layer exists and the given collector-side pn-junction is forward biased. This layer injects holes, which compensate the electrons arriving from exhaustion of the remaining plasma and from dynamic avalanche. This opposes to a removal of the remaining plasma from the collector side. Due to the same polarity of negatively charged acceptor ions in the p-collector and the negative charge of electrons arriving at the collector side, no space charge region can build up at this point even with a high electron density. Therefore, dynamic avalanche of the third degree is not possible in an NPT-IGBT.

Another situation is given in a modern IGBT with a field stop layer or an n-buffer layer in front of the collector layer. This is shown in Fig. 12.42. The electron current flows to the right-hand side. If the hole current coming from the collector is not high enough to compensate the electron current, then at the n^-n-junction a space charge

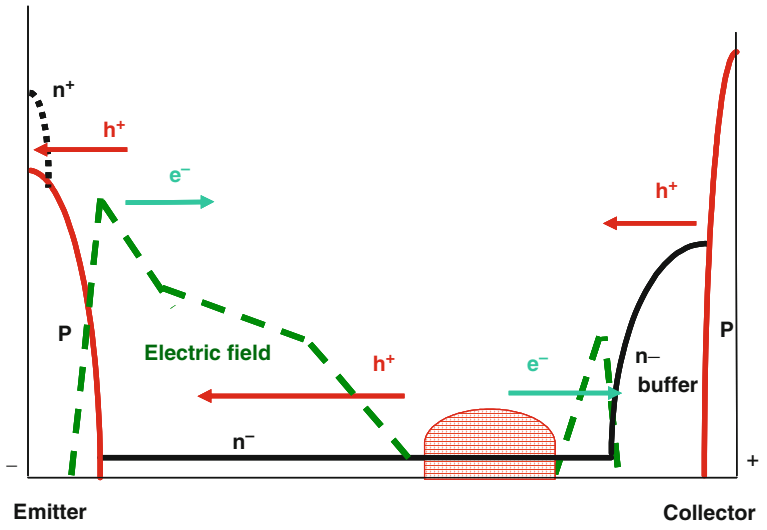


Fig. 12.42 IGBT with an n-buffer at turn-off of an over-current with dynamic avalanche. Under worse conditions a second electric field can rise in front of the n-buffer

can build up between the negative charged free electrons and the positively charged donor ions in the n buffer, and an electric field can rise.

In [Rah05] it is explained that this process is dangerous especially at the final plasma removal and the transition of the IGBT into the "switching self-camping mode" (SSCM, see Sect. 12.4). SSCM occurs in an IGBT during turn-off of an over-current under the condition of a high parasitic circuit inductance. In SSCM a field distribution with a second field peak at the nn^+-junction can occur, similar to second breakdown in a bipolar transistor. This effect is unstable. However, if a sufficiently high hole current is delivered from the p-collector layer and it compensates the electron current, then the SSCM event is stabilized in the IGBT. To achieve this, the emitter efficiency of the p-collector and the corresponding current gain α_{pnp} must not be too small.

Figure 12.43 shows the turn-off of a 3.3 kV-IGBT at a high over-current level. After the voltage has climbed up to approx. 2000 V, a flattening of the voltage course is to be seen. This is a sign of strong dynamic avalanche, where the holes flowing through the space charge limit the blocking capability in this time interval.

After the voltage has reached the applied DC link voltage of 2600 V, the current starts to decay, and still strong dynamic avalanche exists in the device. At a voltage of 3500 V the device transits into the SSCM. The voltage ramps up to 4000 V, which is close to the static avalanche breakdown voltage of the device. The course of the current and voltage waveforms seems to be more stable compared to the SSCM process in a diode (see Fig. 12.21). The hole current, which comes from the p-collector, adds a stabilizing effect during the SSCM [Rah05]. Figure 12.43 shows that very high stress in dynamic avalanche is also possible for IGBTs.

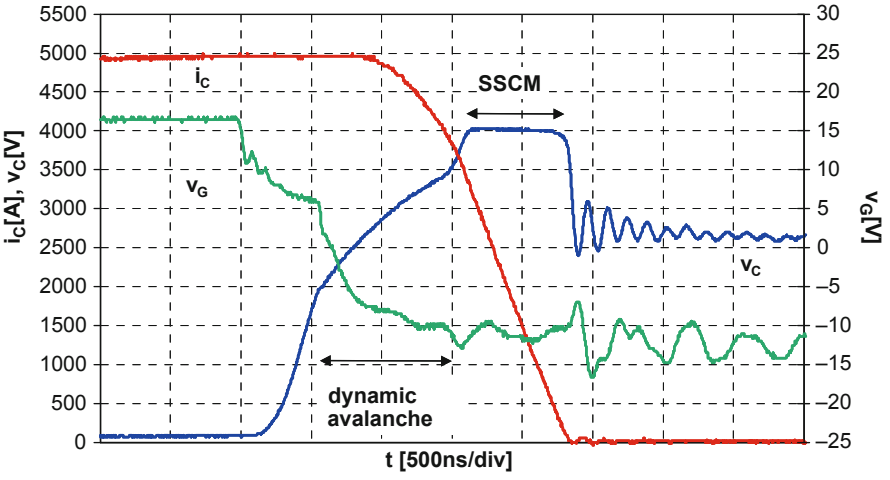


Fig. 12.43 Turn-off of a 3.3 kV 1200 A IGBT module at four times the rated current against a batters voltage $V_{bat} = 2600$ V. After an interval of strong dynamic avalanche, the SSCM event follows. Figure from [Rah05] © 2007 IEEE

12.7 Cosmic Ray Failures

With the introduction of high-voltage semiconductors with turn-off capability in converters for electric traction in the beginning of the 1990s, failures in the application were observed which could not be explained with the available knowledge at the time. The failures occurred during the blocking mode of the devices. The application conditions were employed in the lab and long-term tests with high DC voltage in blocking direction were carried out. The tests confirmed the occurrence of spontaneous failures [Kab94]. The spontaneous character of the failures was strange, since no prior indications in device behavior, e.g., an increase of the leakage current, were found.

Figure 12.44 shows the results of the salt mine experiment. First, six failures occurred in the lab within 700 device hours. The test was interrupted and continued in a salt mine, with 140 m of solid rock above the test position. Under these conditions, no failure occurred. The test in the salt mine was interrupted again and continued in the lab, and now failures occurred once more at a comparable rate as before in the lab. The test position was moved again to a place in the cellar of a multi-storey building, at this position there was in sum 2.5 m of concrete above the test position. The failure rate was reduced again.

With these results, it was proven that cosmic ray was the main reason for failures of power semiconductor devices under such conditions.

The cosmic radiation consists of high-energy particles, which are created in supernova explosions or in the core of distant active galaxies; they hit the earth from all directions. A high-energy primary cosmic ray particle usually does not reach the

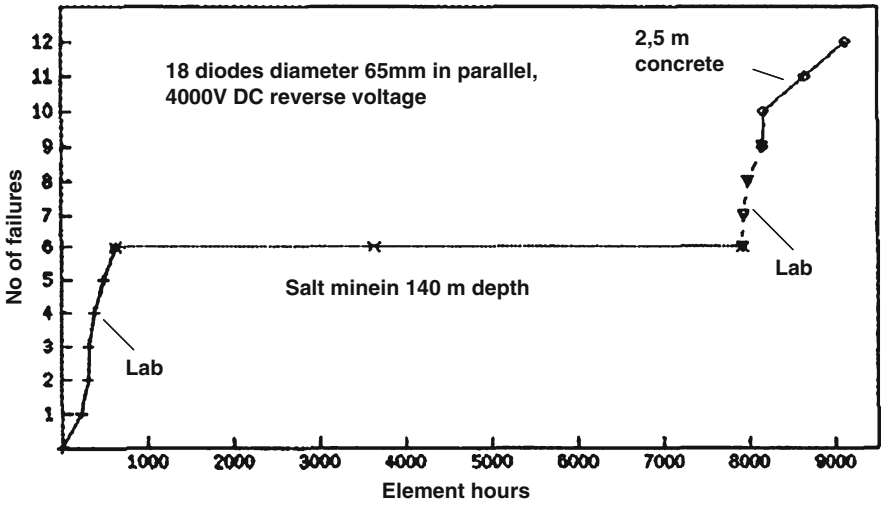


Fig. 12.44 Results of the salt mine experiment. Number of failures over the summarized time of DC voltage stress. Figure from [Kab94] © 1994 IEEE

surface of the earth directly, but collides with atmospheric particles. There it generates a variety of secondary high-energy particles; a cosmic-ray shower arrives at the earth surface as terrestrial cosmic radiation. What is relevant to device failures are high-energy neutrons and protons; however, also the effect of pions and myons is not excluded. A small part of the neutrons, traveling through a semiconductor device, collide with the cores of silicon atoms. Back scattered ions are created, which generate again locally a dense plasma of charge carriers. In the space charge region of a semiconductor device in the blocking mode, these carriers are separated and a current pulse occurs. If the electric field is higher than a certain threshold value – which depends on the initial plasma generation – then impact ionization creates more carriers than the carriers which flow out of the plasma region by the mechanism of diffusion. The discharge runs as a so-called streamer with high velocity through the device, in analogy to a discharge in a gas. The device is flooded locally with free carriers within some hundred picoseconds; hence, a local current tube occurs. Finally, the very high local current density destroys the semiconductor device.

The failed devices exhibit a pinhole-size molten channel from cathode to anode anywhere in the bulk. Failure pictures from device failures caused by cosmic ray are shown in Fig. 12.45. On the left-hand side in Fig. 12.45 a pinhole is to be seen. On the right-hand side bubbles in the metallization are to be seen and below the metallization a pinhole is hidden. The failure pictures show clearly an effect occurring in a very narrow region.

Figure 12.46 shows the cosmic ray failure picture of a 3.3-kV IGBT die. Again a pinhole is found, with a size in the order of the cell pitch of the IGBT die.

Several tests of different device designs were carried out, for acceleration of the failure rate tests stations at high altitudes were arranged (Zugspitze 2964 m, Jungfraujoch 3580 m). Terrestrial cosmic ray increases with altitude above sea level. The acceleration factor amounts to 10 in 3000 m, in 5000 m to approx. 45 [Kai05]. In parallel experiments, tests with particle accelerators were carried out, devices at

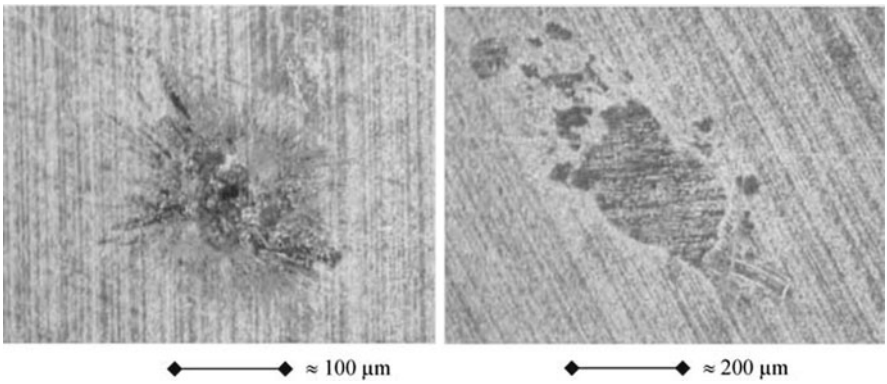


Fig. 12.45 Failure pictures after a cosmic ray destruction of 4.5-kV diodes with a diameter of <50 µm. The photos are from the cathode side. Left-hand side: small pinhole. Right-hand side: Molten area in the metallization with bubbles. Pictures from Jean-Francois Serviere, Alstom

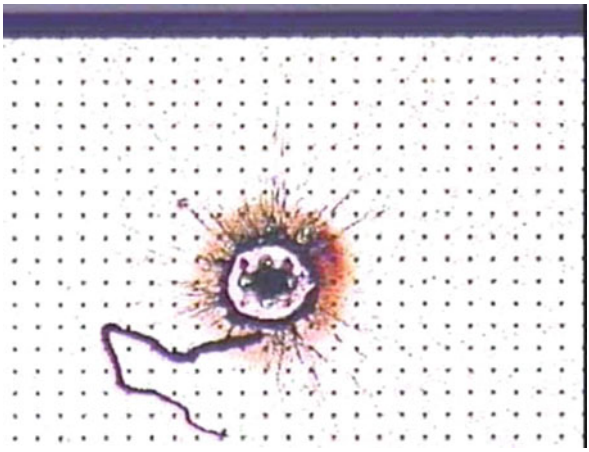


Fig. 12.46 Cosmic ray failure of a 3.3 kV IGBT in the cell area. Cell pitch 15 µm. Picture from G. Sölkner, Infineon

high-reverse voltages were irradiated with high-energy neutrons, protons, and other ions accelerated with high energy.

Irradiation with ions, especially the types of ions and energies which are generated at a neutron–silicon collision, is suited to study the detailed failure mechanism. A great amount of research work was done in this field, see, e.g., [Soe00].

An example of the results is shown in Fig. 12.47, taken from [Kai04]. Diodes rated at 3.3 kV have been irradiated with ^{12}C-ons and the reverse voltage applied during irradiation was increased. Two diode designs were compared having a triangular electric field shapes (NPT, see Fig. 5.3a) and another diode with a trapezoidal shape of the electric field (PT in Fig. 5.3b), which is assigned with FS (field stop). While at a small voltage the charge generated by a single ^{12}C ion is small for all

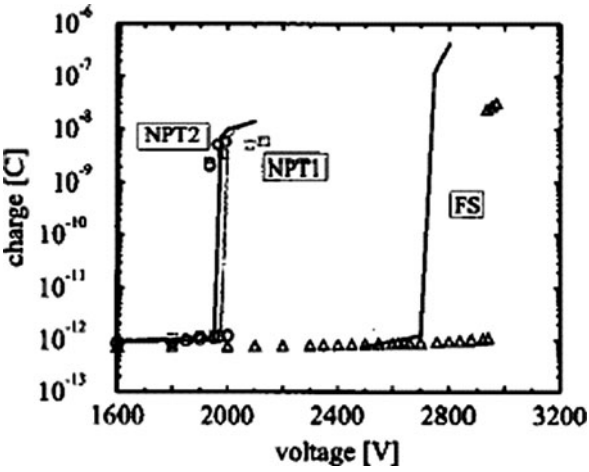


Fig. 12.47 Charge created at irradiation with single ^{12}C-ions of a kinetic energy of 1.7 MeV as a function of the applied reverse DC voltage to a 3.3 kV diode. Simulation (*straight line*) and experiment (*symbols*) for a triangular-filed shape NPT1 (□), NPT2 (O), and a trapezoidal field shape FS (Δ). Figure from [Kai04] © 2004 IEEE

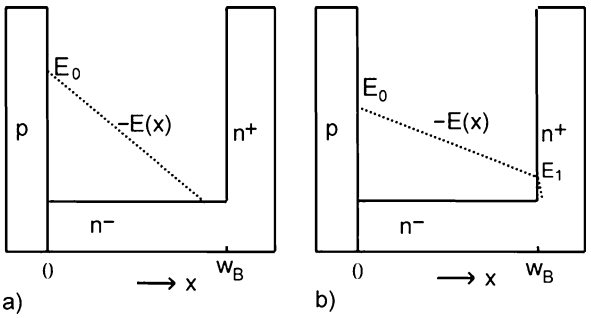


Fig. 12.48 Schematic drawing of the field shape for an NPT design (**a**) and PT design (**b**) at the same thickness and same applied voltage in the blocking direction

samples, above a defined threshold voltage a strong charge carrier multiplication sets on and the created charge increases suddenly more than three decades. For the FS diode, which has a PT design, this threshold voltage is more than 700 V higher when compared to the NPT diode.

Even though PT and NPT dimensioning in Fig. 12.47 are designed for the same rated voltage, the maximal electric field is much lower for the PT design. This is illustrated in Fig. 12.48. The shape of the electric field is drawn for two simplified diode structures with the same thickness at same applied blocking voltage. The area below the line $E(x)$, which corresponds to the reverse voltage, is the same for both devices. However, the value E_0 is much lower for the PT design. At the same voltage, impact ionization is still negligible for the device with PT design. For the occurrence of impact ionization in the PT diode, the applied voltage must be increased. Then the line of $-E(x)$ in Fig. 12.48b is shifted upward, until E_0 becomes close to a value as in Fig. 12.48a. Then impact ionization sets on for the diode with PT dimensioning too; however, at a much higher voltage.

The design of a device for increased cosmic ray stability is in substance done in the way that the maximal occurring DC voltage V_{bat} in the application is used and the value of E_0 is kept as low as possible. For this, the base doping N_D must be made low, so that a PT dimensioning is given. Additionally E_0 is lower, if the device is designed with a wider w_B. However, this is of disadvantage regarding the conduction losses and switching losses.

With these measures, the probability of cosmic ray-induced failures is decreased. However, the occurrence of such failures cannot be avoided completely, since the type and especially the energy of the striking particle cannot be predicted, and also the initial plasma discharge is dependent on some stochastic effects. The cosmic ray stability is specified in "Failure in Time" (FIT), where it holds

$$1\,\text{FIT} = 1\ \text{failure in}\ 10^9\,\text{h}$$

The requirement for a power electronic module for traction application is 100 FIT: One failure is to be expected statistically in 10^7 h, these are 1141 years. One must consider that a typical power module for such an application consists of 24 IGBTs

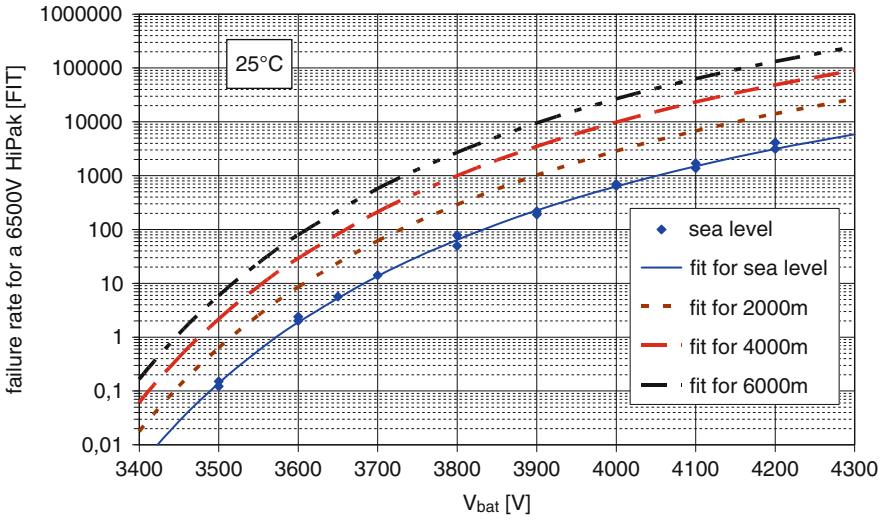


Fig. 12.49 Cosmic ray failure rate at $T = 25°C$ for the 6.5 kV IGBT module 5SNA0600G650100 from ABB. Figure from [Kam04]

and 12 freewheeling diodes. For a single device holds a lower FIT rate, respectively (Fig. 12.49).

The failure rate is highest at low temperatures and is decreasing with temperature due to the decreasing avalanche ionization rates at increased temperature, see Chap. 2.

With the described measures, the given requirements can be fulfilled at specified conditions; however, the detailed final failure mechanism of these "single event burnout" [Alb05] is still a subject of research. In a MOSFET, the failure mechanism is explained to be finally the activation of the parasitic bipolar transistor and second breakdown of the bipolar transistor [Was86]. In diodes, no parasitic component of any transistor type is present. Even strong local avalanche breakdown should be stable, see Sect. 12.4. Some work [Alb05, Kai04, Soe00] explains the final failure mode by very high local heating caused by very strong dynamic avalanche. Field redistribution effects in analogy to dynamic avalanche of the third degree are considered in deep submicron CMOS devices for improved ruggedness against radiation-caused single event pulses [Das07].

For high-voltage devices, the dimensioning rules for cosmic ray stability contradict to other rules for optimizing the device characteristic; e.g. for diodes it contradicts to the requirement for soft-recovery behavior, which is hard to achieve if a strong PT dimensioning is used. A trade-off between different requirements must be made. To meet said demands, most of the nowadays high-voltage devices use designs with a middle layer thickness w_B much higher than necessary for the required blocking capability. However, this leads to increased losses in the forward conduction mode and/or increased turn-off losses.

12.8 Failure Analysis

Some failure mechanisms occurring in power circuits with IGBTs are discussed here in summary and interrelation. Table 12.1 gives different failure mechanisms, which are divided to failures caused by current respectively temperature, by voltage, and by dynamic effects. The failure reasons are printed in cursive letters.

Table 12.1 Some failure mechanisms in IGBT modules. *Cursive: failure reason.* Normal: failure picture

Current temperature	Voltage	Dynamic effects
		Applied voltage below the rated voltage
– *Too high average current* The die shows molten areas with a diameter of several millimeter. Failures located in the active area	– *Production fault* Failure location starts at the edge	– *Lack of dynamic ruggedness of the freewheeling diode* Diode and transistor in the associated commutation loop are destroyed
– *Surge current exceeded* Local molten area, size approx. 1 mm, sometimes cracks in the crystal, failure located in the active area	– *Voltage peaks above rated voltage* Failure location starts at the edge	– *Lack of dynamic ruggedness of the freewheeling diode* Only the diode is destroyed, pinhole with diameter <100 µm
– *Short-circuit capability of IGBT exceeded* Only IGBTs destroyed, large part of the emitter area burned off.	– *Lack of long term stability of the passivation layer* Failure location starts at the edge	– *Dynamic avalanche third degree* Pinhole with a diameter <100 µm, cracks in the crystal lattice originating – *Dynamic latch-up* eventually only 1 IGBT is destroyed

For failures caused by current, a molten zone within the active area of the device is typical. At very high average currents, one finds a destroyed area with a diameter of several millimeters.

If a surge current failure of a diode occurs, the molten area is usually smaller, in the range of 1 mm. For bonded diodes, often a melting of metallization beside the bond feet is observed. Surge current occurs in an application, for example, when a non-loaded DC link capacitor is connected by a diode rectifier circuit to the grid and a very high current pulse is generated in the first instant. In this case, an application fault is given. A loading circuit for the DC-link capacitor will be of help in this case. Manufacturers of power semiconductors know very well the typical surge current failure pictures of their devices, as shown for example in Fig. 12.6, and they can identify such failures.

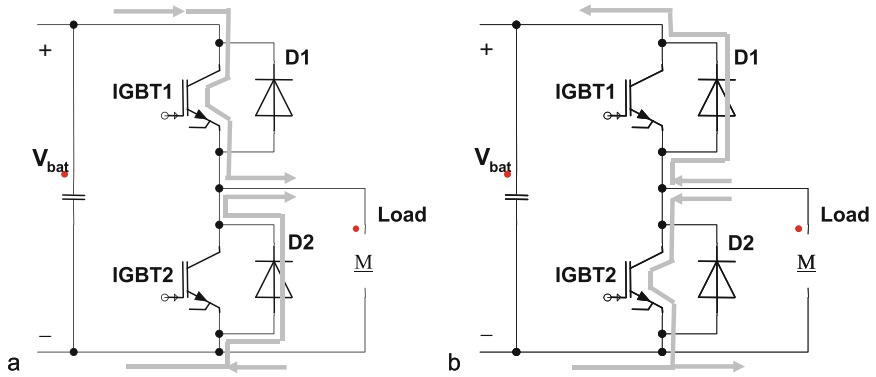


Fig. 12.50 Commutation loops in a half-bridge. (**a**) Power transmission from the DC link to the load. (**b**) Reverse power flux

If failures are caused by voltage, it is mostly observed that the failure location is at the edge of the device, such as in the junction termination structure. At these positions, the highest electric fields occur at the surface. The junction termination is very important in power device manufacturing and is most sensitive to contamination in the production line, to faults in photolithography due to dust particles, etc. If a device has a weak point due to a fault in production, this occurs usually at the edge of the device. While current-induced failures are mostly due to faults in the application, this must not be the case for voltage-induced failures. Application faults – voltage peaks higher than the rated voltage – as well as production faults must be taken into account.

Failures by dynamic effects are mainly related to switching events. The voltage stays below the rated voltage of the device. At switching events the transistors interact with their freewheeling diode. Figure 12.50 shows the corresponding commutation loops. At power transmission to the load (Fig. 12.50a) IGBT1 commutates with diode D2. If diode D2 fails during its turn-off, the associated transistor in the commutation loop turns on to a short circuit; hence, a short circuit within the bridge with very low inductance is given. Therefore, the IGBT may be destroyed by a short circuit. The same holds for the reverse power flux, where the commutation loop for this is shown in Fig. 12.50b. There IGBT2 commutates with diode D1.

If diodes as well as their associated transistors in the commutation loop are destroyed, the failure reason is usually due to the diode. If the diode fails, the IGBT may be destroyed. On the other hand, if the IGBT fails, this event causes usually no stress for the diode and the diode has no reason to fail. An exception might be, if a spark burns inside the module, which destroys further devices.

With such considerations, sometimes conclusions for failure reasons are still possible even if a module is heavily destroyed.

If the freewheeling diode fails and the IGBT turns-off successfully following short circuit, one finds afterward a typical pinhole in the diode. For freewheeling

diodes rated at 1200–1700 V, these pinholes are a sign of lack of dynamic ruggedness. Freewheeling diodes with higher rated voltages, however, may be destroyed by a very high current density in the reverse direction and a contemporaneous high voltage. For such failures by dynamic avalanche of the third degree, one also finds cracks in the crystal lattice, originating at the pinhole. These cracks are a sign of very high local temperature spots. An example was given in Fig. 12.19.

If only the transistor is destroyed, the failure reason must be searched in the transistor. Short-circuit failure is one of the possible mechanisms and is always worth considering. For short-circuit failures, burned off emitter regions with large areas are typical. Note, however, at short circuit III (Sect. 12.6) even a diode failure may occur due to the extreme high dv/dt and voltage spike at diode commutation, while the IGBT may survive [Lut09b].

Another IGBT failure mode is dynamic latch-up, which is caused by single weak cells in the IGBT device. Such failures cannot be found during tests of static parameters at the device manufacturer. For modules with parallel connection of many IGBT dies, the manufacturers execute final tests in an application conform circuit including dynamic turn-off under very high stress conditions in order to find single weak devices and to avoid failures in the targeted application.

Failure analysis is a complex process and a lot of experience is necessary. There are other failure modes, e.g. cosmic ray failures. Also for such failures pinholes are typical. In most cases one cannot conclude a failure reason directly from a failure picture since there are different failure modes which may lead to similar failure pictures. The power circuit and the application condition must be investigated additionally.

With devices connected in parallel, a non-symmetric assembly may lead to oscillations and subsequently to overstress of devices at special positions. Often new questions come forth during failure analysis. In conclusion, failure research is very complex; however, the results are often extremely valuable.

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